

The Porous Decoupling: Vertical Rivalry and Swing-State Leakage

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Abstract

Headline trade statistics show the United States decoupling from China after the 2018 US–China trade war; value-added measures show the opposite—roughly \$22 billion per year in Chinese content still reaches the US through *swing-state* intermediaries (chiefly Vietnam, Mexico, and Malaysia) in critical sectors alone. We explain this porosity as the equilibrium of a contracting problem in which Chinese and non-Chinese intermediaries are complements, so a swing state can formally accept US restrictions yet keep sourcing Chinese inputs whose content is re-exported in finished goods. The model yields two predictions: (i) a cross-sectional gradient—the swing-state premium in embedded Chinese-content share, relative to other major US trading partners, widens in industries facing higher US tariffs on direct Chinese imports; and (ii) a *Paradox of Competitiveness*, in which a more efficient upstream Rival makes voluntary alignment fiscally prohibitive and pushes the US from subsidy to coercion. We confirm (i) in two inter-country input–output databases; the same pattern appears in pre-2018 data, consistent with the model’s steady-state interpretation rather than a 2018 tariff response. US Entity-List actions exhibit the regime-switch signature predicted by (ii): a sharp post-2018 escalation against Chinese firms, set against fifteen years of steady extra-bilateral coercion. Customs-based decoupling metrics therefore systematically overstate real decoupling in exactly the sectors where decoupling is most sought.

Keywords: Geoeconomics, Vertical Supply Chains, Coercion, US-China Rivalry, Entity List.

JEL Classification: F13, F51, L14, O25.

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1 Introduction

Between 2018 and 2024, the United States imposed Section-301 tariffs averaging roughly twenty-five percentage points on Chinese imports concentrated in critical sectors—semiconductors, precision machinery, batteries, and electric-vehicle components. Direct US imports from China in those sectors fell sharply, as policy intended. Over the same window, imports from a small set of swing states—chief among them Vietnam, Mexico, and Malaysia—rose by a comparable amount. The headline change masks a deeper pattern that emerges when we trace the *origin* of the value embodied in these swing-state exports. The input–output accounting literature calls this the *foreign value-added* (FVA) share. Chinese content has continued to reach the US indirectly throughout the Section-301 period, settling near 6% of swing-state critical-sector exports by 2020–2022. That share translates to roughly \$22 billion per year in critical sectors alone.¹ On customs data the US is decoupling; on value-added data it is recoupling through intermediaries. This paper asks why the two measures disagree and what the discrepancy implies for economic statecraft. We answer with a model of an alignment bargain between the US and a swing state and with a single cross-sectional prediction that the value-added data identify directly.

We argue that the discrepancy is not a measurement artifact or a transitional friction. It is the steady-state outcome of a bargain between the US (the downstream *Hegemon* of the supply chain) and a *swing state*. The same production technology that makes the bargain rational also partially voids it in equilibrium. The mechanism rests on complementarity: in critical sectors, non-Chinese intermediates are poor substitutes for Chinese ones. Boehm et al. (2019) and Atalay (2017) document this empirically. A swing state cannot cheaply rebalance its input mix no matter how generously the US pays to align. The unpayable part of the bargain is therefore absorbed by continued sourcing of Chinese inputs, which are then re-exported in finished goods. The same complementarity yields a directly testable cross-sectional prediction. In sectors where US tariffs on direct Chinese imports are higher,

¹Derivation from fitted FVA shares applied to 2022 swing-state-to-US gross exports is given in Section 4.

the Chinese-content share in swing-state exports to the US should rise more steeply than in exports from non-swing-state origins—a steeper tariff gradient for swing states.

We test this prediction in two independently maintained inter-country input–output (ICIO) databases: the OECD ICIO Extended release and the ADB Multi-Regional Input–Output (MRIO) 72-economy release. In a triple-difference specification, we interact published Section-301 tariff rates with swing-state identity. The estimate is positive, statistically robust, and economically meaningful. A 25-percentage-point tariff increase on a swing-state-produced good raises its embedded Chinese-content share by roughly 2.5 percentage points, over and above what industry-by-year and source-by-year fixed effects absorb. The result survives two-way clustering on source and industry, leave-one-out on the swing-state set, and Honest-DiD bounds for parallel-trends violations. Placebo tests recover between 47% and 87% of the coefficient in pre-2018 windows, with the earliest 2010 placebo recovering 87%. We read this not as a concern but as the signature of a steady-state prediction: the pattern is a standing feature of the global production structure, not a quasi-experimental response to the 2018 shock. The policy implication is that customs-based surveillance systematically overstates real decoupling, by an economically meaningful margin, in exactly the sectors where decoupling is most sought.

This measurement gap is the signature of a broader historical shift. When the US sought to align allies in the post-war order, it provided public goods, market access, and subsidies—a generous-empire regime of positive inducements. Between 2018 and 2024, the policy toolkit has pivoted toward negative coercion: Entity Lists, export controls, and the weaponization of market access (Facts 4–6 in Section 2). Swing states caught between the two regimes rely on China for upstream critical inputs while selling finished goods into the US market. Chief among these are South Korea, Taiwan, Vietnam, Malaysia, and Mexico. We term this configuration the *Vertical Trap*. Our model is a three-stage game over this triangular structure (Upstream Rival, Swing State, Downstream Hegemon). The Hegemon’s strategic problem is that eliminating the kill-switch risk requires the Swing State to abandon its most

efficient input supplier. This deviation is costly, and the Hegemon must either subsidize or coerce it.

The model delivers three theoretical results. First, the *Regime Switch* (Proposition 1): the Hegemon’s optimal alignment policy switches regime as the Rival’s productivity θ crosses a critical threshold $\bar{\theta}$. Below $\bar{\theta}$, the regime is Generous Empire ($\lambda^* = 0$); above it, Coercive Hegemon ($\lambda^* > 0$). The equilibrium map $\lambda^*(\theta)$ is continuous but non-differentiable at the kink. Second, the *Paradox of Competitiveness* (Proposition 2): the threshold $\bar{\theta}$ locates the regime switch in the Rival’s efficiency advantage. When the Rival is technologically immature, subsidy-based alignment is fiscally affordable; as the Rival approaches the technological frontier, the subsidy required for voluntary alignment becomes prohibitive, and the Hegemon optimally switches to coercion. The paradox is that the more efficient the Rival becomes, the more coercive the Hegemon must be to maintain its security architecture. Third, the *Leakage Asymmetry* (Propositions 4 and 5): the bottleneck primitive $\sigma < 1$ predicts the cross-sectional pattern identified in the data above—the tariff-by-swing-state interaction effect on Chinese value-added in swing-state $\rightarrow$ US exports. The proposition locates this pattern as a steady-state feature of the production structure rather than a 2018 policy break.

Contribution. The paper sits at five intersections, positioned in detail below. In compact form: to the *geoeconomics of fragmentation*, we endogenize the Hegemon’s choice between inducement and coercion as a function of the Rival’s efficiency on bottleneck inputs. To *common-agency and exclusion contracting*, we show that bottleneck complementarity breaks the truthful-equilibrium benchmark of Bernheim and Whinston (1986) and makes exclusion the rational second-best. To *production networks and supply-chain IO*, we reinterpret a low-substitutability node not as a fragility externality but as a locus of infinite implementation cost for competing principals. To *value-added trade measurement*, we name the cross-sectional pattern most cleanly identified by Leontief-inverse decomposition—the tariff-by-swing-state interaction effect—and test it directly. To the *political economy of policy radicalization*, we model the post-2018 US shift as a regime transition (the kinked alignment correspondence

of Proposition 1) triggered by Rival technology rather than a preference shock.

Scope and what this paper does not do. We take the Rival’s technological rise θ as exogenous and do not model its endogenous accumulation. We treat the security externality λx as reduced-form rather than deriving it from a probabilistic “kill-switch” microfoundation. And we abstract from dynamic reputation or multi-period learning. We microfound the fiscal capacity $\bar{T}$ and the security-policy cost γ in the appendix but take them as primitives in the main model. These limitations are deliberate. The paper isolates a single new mechanism: the endogenous corner solution that emerges when bottleneck complementarity meets a binding fiscal constraint. Dynamic, endogenous-technology, and probabilistic-security extensions are left for future work.

Related literature. Our paper draws on International Trade, Industrial Organization, and Political Economy. First, we contribute to the emerging literature on Geoeconomics and the weaponization of supply chains. The intellectual ancestor of this line is Hirschman (1945), whose analysis of foreign trade as an instrument of national power established the template that modern weaponization scholarship has inherited. The broader international-political-economy tradition on hegemonic ordering and systemic responsibility (Gilpin, 1981) provides the historical framing within which our model’s subsidy-vs-coercion regime switch can be read. Farrell and Newman (2019) revived this tradition by introducing “Weaponized Interdependence,” arguing that asymmetric hub-and-spoke networks allow central nodes to coerce peripheral actors. Farrell and Newman (2023) and the essays in Drezner et al. (2021) extend this to financial and digital infrastructures. Clayton et al. (2024) formalize this in general equilibrium by modeling a “Hegemon’s Dilemma” between coercive power and economic integration. Their dyadic framework is the most general existing treatment of geoeconomic bargaining. Clayton et al. (2026) extend this to a three-player setting with a threshold at which the Hegemon switches between inducement and coercion; that threshold is a preference parameter in their formulation. We build on their insight that coercion has

an endogenous cost. We specialize the mechanism to a triangular (U–S–C) supply chain, in which a production-structure primitive ($\sigma_c < 1$) generates leakage that partially voids the Hegemon’s contract and pins the regime-switch threshold to production technology rather than preferences. Policy-oriented complements cover shifting trade linkages under geopolitical tension (Gopinath et al., 2024), the erosion of multilateral trade rules (Mattoo et al., 2024), and the costs of geoeconomic fragmentation (Aiyar et al., 2023). Sanctions economics (e.g., Mulder, 2022; Itskhoki and Mukhin, 2025) characterizes the macro consequences of coercion once deployed. These models typically treat the Hegemon’s coercive capacity as a static endowment or reduced-form shock. Our contribution is to endogenize the *decision* to coerce. Lim (2015) classifies coercion strategies (sticky vs. slippery power), but does not microfound when the Hegemon switches between them. We provide that microfoundation: the switch from positive inducements (aid/subsidies) to negative coercion (sanctions) occurs precisely when the Rival becomes efficient on a bottleneck input.

Second, we differentiate our work from the horizontal strategic-trade-policy tradition (Brander and Spencer, 1985; Eaton and Grossman, 1986), in which nations compete to sell substitute goods and subsidy wars are the dominant conflict. We build instead on the vertical industrial-organization and global-value-chain literature (Yi, 2003; Antràs, 2003; Grossman and Helpman, 2002; Antràs and Chor, 2013; Alfaro et al., 2019; Antràs, 2016). We also draw on the quantitative trade and China-shock empirics on tariff incidence and supply-chain reallocation (Autor et al., 2013; Caliendo and Parro, 2015; Fajgelbaum et al., 2020; Amiti et al., 2019; Flaaen et al., 2020; Grossman et al., 2024; Alfaro and Chor, 2023). Our “Bottleneck” assumption is grounded in network-IO work showing that cascading failures and coercion are most effective at low-substitutability nodes (Acemoglu et al., 2012; Elliott et al., 2022; Baqaee and Farhi, 2019, 2024; Carvalho et al., 2021; Boehm et al., 2019). It is further grounded in estimated elasticity ranges from Broda and Weinstein (2006) and Caliendo and Parro (2015). We reinterpret these nodes not merely as fragility points but as loci of *infinite implementation cost* for competing principals. Relative to the industrial-

policy revival (Juhász, 2018; Liu, 2019; Lashkaripour and Lugovskyy, 2023; Juhász et al., 2024; Bown, 2023, 2024) and the sector-specific semiconductor supply-chain literature (Khan et al., 2021), we provide the complementary theoretical mechanism that rationalizes why coercion (rather than subsidy alone) becomes optimal beyond a critical efficiency threshold.

Third, methodologically, we model the US–China rivalry as a Common Agency Game in which two principals bid for an agent (the Swing State). Bernheim and Whinston (1986) and Dixit (1996) established the “Menu Auction” benchmark in which truthful equilibria maximize joint surplus; Grossman and Hart (1983) supplies the underlying principal-agent structure. Standard common-agency models assume linear transfer technology (cash is a perfect substitute for the agent’s action). We introduce a non-linear friction: the technological bottleneck. Following Martimort (1996), Segal (1999), and McAfee and Schwartz (1994), we show that when one principal controls an essential complement ($\sigma < 1$) the Rival cannot compete via linear transfers and must resort to breaking the interaction—exclusion. This mirrors the exclusive-contracting and foreclosure literature (Rasmusen et al., 1991; Segal and Whinston, 2000b,a; Jehiel et al., 1996; Rey and Tirole, 2007), applied here to sovereign states. Our treatment of (R, λ) as an announced contract also connects to incomplete-contracting and authority frameworks (Aghion and Tirole, 1997; Maskin and Tirole, 1999), and to the theory of trade agreements as commitment devices (Bagwell and Staiger, 1999; Maggi and Rodríguez-Clare, 1998; Antràs and Staiger, 2012).

Fourth, our empirical strategy builds on the *value-added trade* measurement literature. The Koopman–Wang–Wei decomposition (Koopman et al., 2014) and the bilateral value-added exports framework of Johnson and Noguera (2012) provide the accounting identities underlying our FVA-China share variable. Wang et al. (2013) extend the KWW decomposition to bilateral-sector levels, which is the exact accounting object entering our panel. Borin and Mancini (2019) map the competing decomposition conventions to the economic questions they are best suited to address, and we follow their guidance in selecting the forward-looking FVA variant for the leakage test. The upstreamness/downstreamness measures of Antràs and

Chor (2018) formalize the positional vocabulary (upstream intermediate, downstream final) that our triangular U–S–C structure invokes. These measures also provide the accounting basis for distinguishing direct from indirect Chinese value-added embodied in swing-state exports. We use this methodology as implemented in the OECD Trade in Value Added (TiVA) and ADB MRIO databases. On the empirical-rerouting side, our paper complements a growing body of descriptive evidence documenting trade diversion through third countries following the 2018 tariffs. Mattoo et al. (2023) show that roughly one-third of the observed decline in US–China bilateral trade reflects rerouting through Southeast Asian and Latin American intermediaries rather than genuine supply-chain restructuring. Alfaro and Chor (2023) document the “great reallocation” of global value chain (GVC) activity from China to ASEAN economies. Schulze and Xin (2025) distinguish trade *reallocation* (domestic substitution for Chinese exports) from trade *rerouting* (transshipment) in six Asian connector countries. Our FVA cross-partial is designed to measure this distinction directly. Ahn and Tan (2025) quantify the welfare gains from supply-chain diversification in a multi-country IO model, a complementary framing to our triangular alignment-contract approach. We go beyond description: the alignment-contract model provides a structural explanation for *why* rerouting concentrates in critical sectors (Proposition 4) and *why* it is steeper for swing-state exporters with higher baseline Chinese-intermediate dependence (Proposition 5).

Finally, we contribute to the political economy of policy radicalization. Most models of trade war treat tariffs as the outcome of distributional conflict (Grossman and Helpman, 1994) or bargaining failure (Bagwell and Staiger, 1999; Maggi and Rodríguez-Clare, 1998; Handley and Limão, 2017), and a growing empirical literature links trade shocks to political backlash and the erosion of centrist coalitions (Autor et al., 2020; Rodrik, 2018). We instead model the shift to coercion as a regime transition. A continuous change in Rival technology (θ) triggers a qualitative switch in the structure of the optimal policy (λ)—from a corner solution at $\lambda^* = 0$ to an interior solution at $\lambda^* > 0$. This switch is orthogonal to (but may be reinforced by) the distributional channel. The equilibrium map $\lambda^*(\theta)$ is continuous but

non-differentiable at the threshold (a kink, not a jump). Empirically, this is consistent with the structural break in US trade and industrial policy documented after 2018 (Fajgelbaum et al., 2020; Amiti et al., 2019; Flaaen and Pierce, 2024; Bown, 2024; Alfaro and Chor, 2023). We show that the “Hawkish” turn in US policy is not a preference shock but a structural necessity—a corner solution that emerges when the interior (subsidy-based) solution becomes fiscally unbounded. These five literatures frame the contribution; the six stylized facts that follow discipline the model’s primitives and motivate the theoretical setup in Section 3.

2 Empirical Patterns

Six stylized facts motivate our theoretical framework and characterize the evolving structure of the global semiconductor and high-tech supply chain. Facts 1–3 (Section 2.1) document trade-flow patterns; Facts 4–6 (Section 2.2) document the US policy response. Together these patterns highlight the structural asymmetry between upstream reliance on Chinese inputs and downstream reliance on the US market, creating what we term a “Vertical Trap” for third-party countries (Swing States). Section 4 returns to the data after the model and formally tests the core claim that leakage is a steady-state feature of the production structure rather than a 2018 policy break.

2.1 Trade Patterns: Vertical Dependence and Indirect Linking

The first set of facts establishes the economic constraints facing swing states. We analyze aggregate trade flows for non-US, non-China economies using UN Comtrade data from 2000 to 2024.

Fact 1: The Vertical Trap. Contrary to the narrative of global fragmentation, swing states have become structurally more integrated with the Chinese supply chain. Figure 1 plots the aggregate trade shares of swing states. The red line shows the share of imports sourced from China (upstream dependence); the blue line shows the share of exports destined

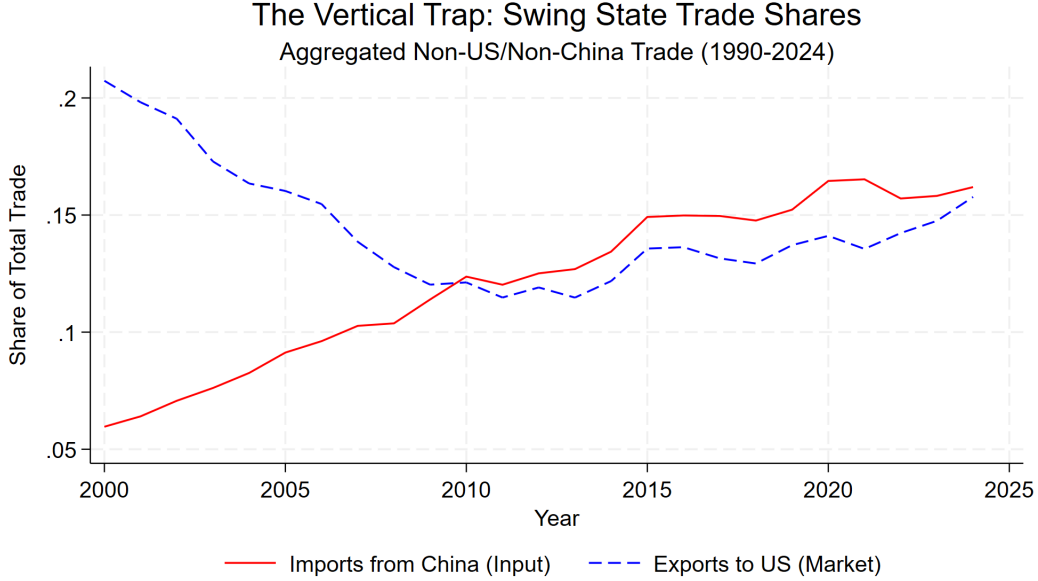


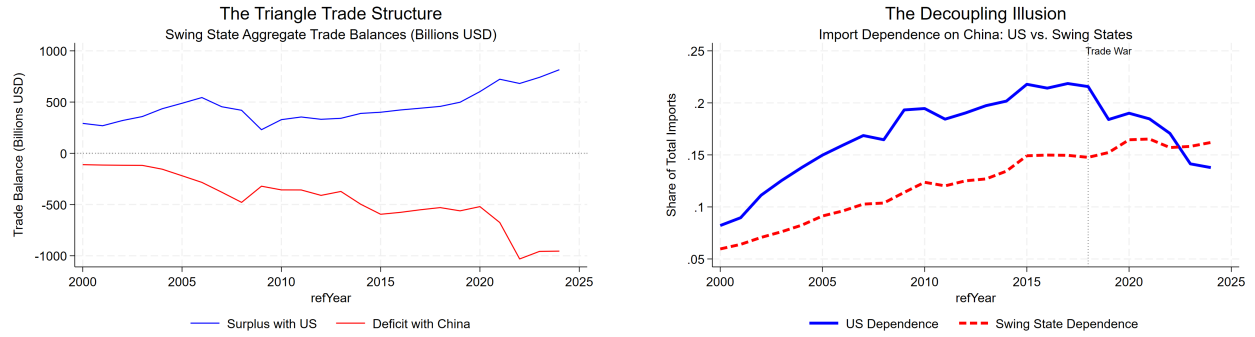
Figure 1: **The Vertical Trap.** Aggregated trade shares of swing states (non-US/non-China). The reliance on Chinese inputs (Red) has structurally surpassed the reliance on the US export market (Blue), creating an upstream bottleneck.

for the US (downstream dependence). Crucially, we observe a “crossover” around 2010. While the US remains a critical export market (absorbing $\sim 15\%$ of exports), the reliance on Chinese inputs has tripled, rising from 6% to over 16%. This confirms the model’s assumption that China dominates the upstream bottleneck ($\sigma < 1$), anchoring swing states to the “Red Supply Chain” despite geopolitical pressure.²

Fact 2: Triangle Trade and Indirect Linking. How do swing states reconcile this upstream dependence with US pressure to decouple? The data suggest a mechanism of “Triangle Trade,” where swing states act as processing hubs. Figure 2(a) reveals a striking symmetry. As the swing states’ aggregate trade surplus with the US expands to \$800 billion, their trade deficit with China deepens proportionally to \$1 trillion. This mirror image implies a pass-through structure: Chinese inputs are imported, processed, and re-exported to the US.

²The same pattern appears in IMF Direction of Trade Statistics (DOTS) bilateral trade shares. It is also documented in the trade-reallocation evidence of International Monetary Fund (2023, Chapter 4).

Fact 3: The Decoupling Divergence. This structural integration limits the efficacy of US bilateral tariffs. Figure 2(b) compares the import dependence of the US versus swing states. Following the 2018 Trade War (dotted line), US direct reliance on China (Solid Blue) dropped significantly. However, swing-state reliance on China (Dashed Red) did not decline; it stabilized and remained high. This divergence provides empirical support for our “Leakage” hypothesis: coercive policies successfully severed direct US-China links but failed to break the indirect links embedded in the swing-state supply chain.



(a) **The Triangle Trade.** Symmetrical deepening of the US surplus and China deficit.

(b) **The Decoupling Illusion.** Swing-state reliance on China resists US policy shocks.

Figure 2: **Mechanisms of Evasion.** Panel (a) shows the flow of value from China to the US via swing states. Panel (b) shows that while the US decoupled directly, the swing states did not, rendering the decoupling porous.

2.2 Policy Responses: The Expansion of the Coercive Frontier

In response to the “leakage” observed in trade flows, US policy has undergone a structural transformation. We analyze the evolution of the US Bureau of Industry and Security (BIS) Entity List to document the shift from bilateral trade disputes to a systemic network blockade.

Fact 4: From Bilateral to Network Coercion. A salient stylized fact is that the coercive perimeter has been extra-bilateral for at least fifteen years. The stock of third-country listings has exceeded the stock of direct China listings in every year since the mid-2000s, with ratios ranging from roughly 5:1 in 2010 to 9:1 in 2017. Figure 3 plots the

cumulative count of active entities on the Entity List on a log scale, which makes growth rates rather than absolute levels readable. Two features of the plot matter. First, the Third-country line (solid blue) has a roughly constant log-slope throughout the sample—its compound growth rate is $\approx 23\%$ per year in 2010–2017 and $\approx 19\%$ per year in 2018–2024, essentially unchanged by Section-301. Second, the China line (dashed red) is nearly flat from 2005 to 2017 ($\approx 13\%$ per year) and then kinks sharply upward after 2018 ($\approx 41\%$ per year). Post-2018 the bilateral stock converges toward the third-country stock, narrowing the level ratio to roughly 3:1 by 2024. The *infrastructure* for third-country coercion was therefore already in place before the tariff shock; what 2018 triggered was a bilateral intensification *on top of* a pre-existing extra-bilateral coercive machine. Both patterns match the model. Once the Rival (China) dominates the upstream bottleneck, the Hegemon (US) maintains and escalates coercion both against the Rival directly and against the global intermediary network, including nominal swing states. The third-country machinery is the steady state, not a 2018 innovation. We do not claim a structural break test; the description is observational.

Facts 5 & 6: Precision targeting and evasion resistance. Two further qualitative shifts are consistent with the coercion mechanism, though they do not identify it. First, the Entity List’s sectoral focus has narrowed toward high-technology targets: the share of new entries containing “Semiconductor,” “AI,” or “Quantum” rose from under 10% in the early 2010s to over 20% by 2024. Second, administrative effort per listing has risen. The average number of aliases per sanctioned entity spiked to nearly 2.7 in 2023, a rough proxy for the “Cat-and-Mouse” dynamic that Section 3 formalizes via the efficiency gap ΔC . We report the underlying series in Appendix Figure 11; these patterns motivate but are not load-bearing for the formal analysis.

Facts 1–6 document aggregate co-movements consistent with the leakage mechanism but do not test it. They motivate three structural primitives that Section 3 formalizes. First, bottleneck complementarity ($\sigma_c < 1$) between upstream inputs in critical sectors (Facts 1–2).

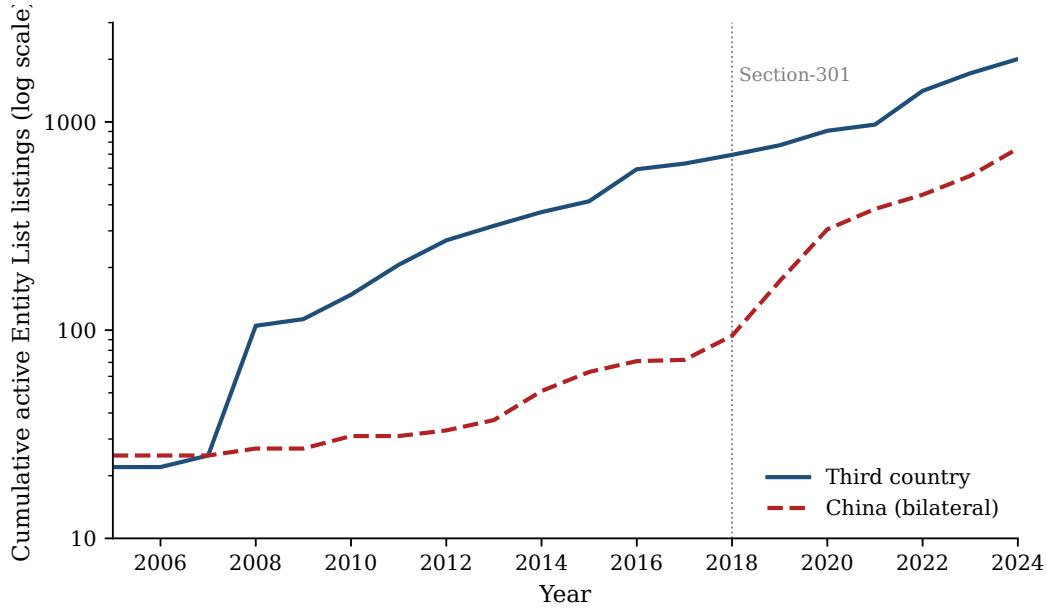


Figure 3: **The Network War.** Cumulative count of active US Entity List listings on a log scale, split by country of the listed entity (third-country vs. China-bilateral). The third-country line has a roughly constant log-slope throughout, indicating steady compound growth ($\approx 23\%$ p.a. 2010–2017 and $\approx 19\%$ p.a. 2018–2024); the China line is nearly flat pre-2018 ($\approx 13\%$ p.a.) and kinks upward post-2018 ($\approx 41\%$ p.a.). The extra-bilateral coercive infrastructure preceded Section-301; 2018 added a bilateral intensification on top of it.

Second, a strategic alignment margin on which swing states balance Chinese input efficiency against US market access (Fact 3). Third, an endogenous coercive response by the Hegemon once the alignment bargain turns fiscally unbounded (Facts 4–6). Section 3 builds a three-stage model of that alignment bargain and derives two testable predictions—the flow-margin leakage of Proposition 4 and the steady-state cross-partial of Proposition 5. Section 4 then returns to the data with regression tests of both predictions.

3 Model

We develop a three-stage vertical game solved by backward induction. Section 3.1 fixes the environment. Sections 3.2–3.4 derive each stage’s best response. Section 3.5 then establishes the *Paradox of Competitiveness*: an exogenous rise in upstream efficiency triggers downstream coercion in equilibrium. Section 3.6 quantifies the resulting welfare loss. Sec-

tion 3.7 derives the multi-sector leakage asymmetry that anchors the empirical cross-partial in Section 4. Section 3.8 closes with three extensions that do not alter the core results.

3.1 The Model Environment

We model a global economy characterized by Vertical Geoeconomic Rivalry. Unlike standard trade models that depict competition between symmetric nations, we treat the global order as a vertical supply chain in which power derives from position: control over upstream inputs versus control over downstream market access.

3.1.1 Agents and Vertical Structure

The economy consists of three strategic players arranged hierarchically. First, **Hegemon C (Upstream)**—the *Rival* (e.g., China)—acts as a monopolist supplier of a critical intermediate input, with the input price w_C as its strategic instrument. Second, **Hegemon U (Downstream)**—simply the *Hegemon* in the main text (e.g., the USA)—acts as a monopolist buyer of final goods and provider of essential intellectual property; its strategic instruments are the market access schedule $R(x)^3$ and the strategic doctrine λ . Finally, **The Swing State (Midstream)** is a representative firm or sector k (e.g., South Korea) that transforms intermediate inputs into final goods. It chooses the alignment $x \in [0, 1]$, which represents the degree of supply chain decoupling.

³The schedule $R(x)$ is a *contingent contract*, not a linear spot price: Hegemon U commits to pay $R(x)$ to the Swing State conditional on realized alignment. This structure answers two parallel questions that arise once the stage objectives are written down. (i) *Why does the Swing State’s revenue depend on its alignment x* (Stage 3, §3.2)? Because $R(\cdot)$ is contractually conditioned on x . The Swing State’s objective $\Pi(x) = R(x) - C(x; w_C)$ is a primitive of the principal–agent problem we model, not the reduced form of a firm choosing output under a market price; the Swing State is an institutional entity that chooses the alignment *policy* x , and $C(x; w_C)$ inherits the CES input-mix *shape* from domestic production at alignment x without x being optimized out of it. (ii) *Why does Hegemon U’s economic welfare take the form $V_{US}(x) - R(x)$* (Stage 2, §3.3)? Because $V_{US}(x)$ is Hegemon U’s gross benefit from Swing-State delivery at alignment x , and $R(x)$ is the contractual transfer paid for that delivery—the two sides of one bilateral ledger. Appendix A.4 derives the reduced form $R(x) = C(x; w_C) + V_{out}$ from the binding participation constraint and documents concrete implementations (preferential tariff, CHIPS-Act-style conditional subsidy, defense-coproduction access).

3.1.2 Timing of Events

The interaction unfolds in three stages, assuming a Stackelberg leadership structure that reflects the rigidity of global supply chains. In **Stage 1**, Hegemon C sets the export price of critical inputs w_C to maximize a weighted sum of commercial profits and geopolitical influence. In **Stage 2**, observing w_C , Hegemon U determines its strategic doctrine λ (the weight placed on security) and announces a market access schedule $R(x)$ to induce alignment. Finally, in **Stage 3**, the Swing State observes the input prices (Chinese input w_C and the Alternative-supplier price w_A of non-Chinese inputs) and Hegemon U's policy schedule $R(x)$, then chooses the alignment x to maximize profit.

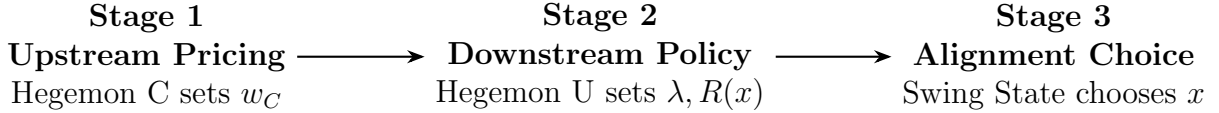


Figure 4: The Timeline of Vertical Rivalry

Assumption 1 (Commitment and Information). *The extensive form in Figure 4 operates under two informational and institutional restrictions. (i) Sequential commitment: At each stage, the acting player commits to their strategic variable before the next player moves. In particular, Hegemon U commits to the pair $(\lambda, R(\cdot))$ prior to the Swing State's alignment choice, and this commitment is binding within the game. (ii) Complete information: All structural parameters are common knowledge. These include the input prices, the production-technology primitives introduced in §3.1, and the fiscal-cost and outside-option parameters introduced with Hegemon U's and the Swing State's problems below.*

Discussion. Assumption 1 plays a central role: the credibility of ex-ante commitments in geoeconomics—sanctions threats, security guarantees, access regimes—is itself an institutional achievement, not an automatic feature of the extensive form. We take it as primitive; its shadow cost is formalized when Hegemon U's problem is specified in §3.3.

3.1.3 The Swing State’s Supply Chain

The Swing State sources inputs from two distinct pools: inputs from the Upstream Hegemon (I_C , the “Red Supply Chain”) and inputs from alternative suppliers (I_A , the “Blue Supply Chain”⁴). The Swing State chooses a geopolitical alignment $x \in [0, 1]$, defined as the physical share of inputs sourced from the Alternative supply:

$$I_A = xI \quad \text{and} \quad I_C = (1 - x)I, \quad (1)$$

where I is the total aggregate input bundle. x is a *policy-imposed physical allocation share*, not an input choice solving a cost-minimization problem. Let w_C and w_A denote the prices of Chinese and Alternative inputs, respectively. We assume $w_C < w_A$, reflecting the Upstream Hegemon’s comparative advantage or subsidization.

3.1.4 The Decoupling Cost

We model the Swing State’s *Decoupling Cost* as a function $C(x; w_C) : [0, 1] \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+$ that measures the resource penalty the Swing State incurs for operating its supply chain at alignment x given an input price w_C charged by the Upstream Hegemon. $C(x; w_C)$ is a primitive object of the model: x enters C as a *policy parameter* (the alignment-imposed input mix), not as a control variable of a cost-minimization problem, consistent with the principal–agent framing developed in §3.2 and formalized in Appendix A.4. The firm-theoretic benchmark, against which we compare the Swing State’s contracted alignment, is the laissez-faire share $x^{\text{LF}}(w_C)$ a cost-minimizing agent would select.

We impose the following minimal regularity conditions on C :

⁴The “Blue Supply Chain” denotes the non-Chinese, non-US input pool the Swing State can source from—principally Japanese, European, Korean, and Taiwanese producers in semiconductor and advanced-electronics supply chains. We exclude Hegemon U from this pool by construction: including US input producers would open a direct input-sales revenue channel from I_A into $V_{US}(x)$, a general-equilibrium spillover that the partial-equilibrium Stage-2 analysis abstracts from. w_A is the effective unit price the Swing State faces when sourcing from this non-US, non-Chinese pool.

Assumption 2 (Decoupling Cost: Bottleneck Regularity). *The Decoupling Cost $C(x; w_C)$ is twice continuously differentiable on $(0, 1) \times \mathbb{R}_{++}$ and satisfies, for every $w_C \in \mathbb{R}_{++}$:*

(C1) *Laissez-faire minimum. There exists a unique $x^{\text{LF}}(w_C) \in (0, 1)$ at which $C'(x^{\text{LF}}; w_C) = 0$; this is the cost-minimizing benchmark a firm-theoretic agent would select.*

(C2) *Strict convexity on the decoupling range. $C''(x; w_C) > 0$ for all $x \in (x^{\text{LF}}(w_C), 1)$.*

(C3) *Explosive marginal decoupling cost. $\lim_{x \rightarrow 1^-} C'(x; w_C) = +\infty$.*

(C4) *Chinese-price sensitivity of the gradient. $\partial C'(x; w_C) / \partial w_C < 0$ for all $x \in (x^{\text{LF}}(w_C), 1)$.*

Condition (C1) identifies the laissez-faire reference point. Conditions (C2)–(C3) are the *bottleneck*: beyond the laissez-faire mix, marginal decoupling becomes strictly costly and prohibitively so as the Swing State approaches full decoupling. Condition (C4) is the key comparative-statics primitive: a fall in w_C (a cheaper Chinese input) steepens the marginal Decoupling Cost on the decoupling range, capturing the model’s central economic insight that “China’s Rise makes the Swing State harder to align.” Together (C1)–(C4) describe a chokepoint technology — rare-earth processing, legacy semiconductors, battery-grade precursors — where the Rival’s upstream input cannot be substituted away cheaply. Except where noted, the analysis that follows uses only Assumption 2.

Canonical example: a CES supply chain. A concrete parameterization that satisfies Assumption 2 arises when the Swing State’s technology aggregates inputs through a Constant Elasticity of Substitution (CES) form with elasticity σ , distribution parameter $\alpha \in (0, 1)$, and $\sigma < 1$ (gross complements). Applying Shephard’s lemma to the CES aggregator evaluated at the alignment mix $(1 - x, x)$ yields the closed form

$$C(x; w_C) = \frac{(1 - x) w_C + x w_A}{\left[\alpha (1 - x)^{\frac{\sigma-1}{\sigma}} + (1 - \alpha) x^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}}, \quad (2)$$

with laissez-faire share

$$x^{\text{LF}}(w_C) = \frac{(1 - \alpha)^\sigma w_C^\sigma}{\alpha^\sigma w_A^\sigma + (1 - \alpha)^\sigma w_C^\sigma}.$$

Under $\sigma < 1$ and $w_C < w_A$, properties (C1)–(C4) follow by direct computation (see the proof of Lemma 1 below). We invoke this CES parameterization in §5 for the quantitative illustration and in §3.8 for the empirical defense of the bottleneck restriction, but not elsewhere: all analytic results use Assumption 2 alone.

3.2 Stage 3 (Swing State): Alignment Choice

We solve the model via backward induction, beginning with the final stage. The Swing State (Agent) takes the input prices $\{w_C, w_A\}$ and the Downstream Hegemon’s policy schedule $R(x)$ as given. It chooses the alignment $x \in [0, 1]$ to maximize its total profit.

Outside option. If the Swing State rejects Hegemon U’s contract, it reverts to the default arrangement—sourcing inputs from the Rival (“integration with China”) with no US-granted market access. Let V_{out} denote the Swing State’s reservation profit under this fallback; we normalize $V_{out} = 0$ throughout, since an additive constant in the outside option shifts the policy schedule $R(x)$ one-for-one and is comparative-statically irrelevant. The Swing State accepts Hegemon U’s contract only when its optimized profit weakly exceeds V_{out} .

Profit maximization. The profit function $\Pi(x)$ is defined as the market access revenue minus the Decoupling Cost:

$$\max_{x \in [0, 1]} \Pi(x) = R(x) - C(x; w_C) \tag{3}$$

Assuming the policy schedule $R(x)$ is differentiable, the optimal alignment x^* is characterized by the First Order Condition:

$$R'(x^*) = C'(x^*) \quad (\text{Incentive Compatibility}) \tag{4}$$

This condition implies that to induce any non-zero alignment $x^* > 0$, the Downstream Hegemon must provide a marginal market access benefit exactly equal to the marginal cost of decoupling.

Properties of the Decoupling Cost. The behavior of the Swing State is driven by the shape of the Decoupling Cost $C(x; w_C)$ on the decoupling range $(x^{\text{LF}}(w_C), 1)$ where the principal's contract pushes alignment above the laissez-faire benchmark.

Lemma 1 (The Bottleneck Gradient). *Under Assumption 2, for every w_C and every $x \in (x^{\text{LF}}(w_C), 1)$:*

1. *Monotonicity.* $C'(x; w_C) > 0$; decoupling beyond the laissez-faire benchmark always incurs a strictly positive marginal cost.
2. *Explosive marginal cost.* $\lim_{x \rightarrow 1^-} C'(x; w_C) = +\infty$; fully replacing the upstream input is asymptotically infeasible.
3. *Input-price sensitivity.* A fall in the upstream price steepens the Decoupling-Cost gradient: $\partial C'(x; w_C) / \partial w_C < 0$.

Proof. All three claims are immediate consequences of Assumption 2. Monotonicity follows from (C1)–(C2): $C'(x^{\text{LF}}; w_C) = 0$ and C is strictly convex on $(x^{\text{LF}}, 1)$, so $C'(x; w_C) > 0$ on that range. Explosive marginal cost is (C3). Input-price sensitivity is (C4). $\square$

CES verification. For the CES parameterization (2), direct differentiation gives $C'(x; w_C) = (w_A - w_C)/A(x) - [(1 - x)w_C + xw_A] \cdot A'(x)/A(x)^2$, where $A(x) = [\alpha(1 - x)^{(\sigma-1)/\sigma} + (1 - \alpha)x^{(\sigma-1)/\sigma}]^{\sigma/(\sigma-1)}$. Under $\sigma < 1$ and $w_C < w_A$: (C1) is the interior first-order condition yielding $x^{\text{LF}}(w_C)$; (C2) follows because the CES aggregator under $\sigma < 1$ is strictly concave in x on $(x^{\text{LF}}, 1)$, so the unit-cost wedge generates strictly convex C ; (C3) follows because $A(x) \rightarrow 0$ as $x \rightarrow 1^-$ (the $\alpha(1 - x)^{(\sigma-1)/\sigma}$ term diverges with exponent $(\sigma - 1)/\sigma < 0$, while the aggregator exponent $\sigma/(\sigma - 1) < 0$ sends A itself to zero),

so C and C' diverge; (C4) follows because $\partial C/\partial w_C = (1 - x)/A(x)$ is larger at low x than at high x , so a fall in w_C reduces the cost of low- x operating points by more than high- x ones, steepening the marginal Decoupling Cost on the decoupling range.

Strategic Implication: The lemma implies that “China’s Rise” ($w_C \downarrow$) does not merely make the Swing State richer; it makes the Swing State *harder to align*. The Downstream Hegemon faces a steeper marginal cost curve $C'(x)$ to implement the same target x^* , requiring a more aggressive (steeper) policy schedule $R(x)$.

3.3 Stage 2 (Downstream Hegemon): Policy and Doctrine

In Stage 2, the Downstream Hegemon (US) designs the policy schedule $R(x)$ and selects its strategic doctrine λ to maximize national welfare, anticipating the Swing State’s reaction in Stage 3.

Joint objective function. The Hegemon’s utility $\mathcal{W}_{US}$ is defined over the realized alignment x , the transfer payment $R(x)$, and the security weight $\lambda \in [0, 1]$. We model the choice of λ not as a fixed preference, but as an endogenous policy decision (“Securitization”) that incurs convex institutional costs:

$$\mathcal{W}_{US}(x, R(x), \lambda) = \underbrace{\lambda x \theta(w_C)}_{\text{Security Utility}} + \underbrace{(1 - \lambda)[V_{US}(x) - R(x)]}_{\text{Net Economic Surplus}} - \underbrace{\frac{\gamma}{2}\lambda^2}_{\text{Cost of Securitization}} \quad (5)$$

where $V_{US}(x)$ represents the gross economic value generated for the US, such as consumer surplus or IP royalties. We assume $V'_{US}(x) < 0$, reflecting the efficiency loss from decoupling from the most efficient upstream supplier. The parameter $\gamma > 0$ quantifies the institutional resistance to adopting a security-heavy doctrine—fiscal constraints, domestic political polarization, or administrative capacity. Equivalently, γ is the reduced form of the commitment cost implicit in Assumption 1: credibly binding to a deviation from the default market-integration doctrine is itself costly (in the spirit of Rogoff, 1985; Maskin and Tirole, 1999;

Aghion and Tirole, 1997).

The multiplier $\theta(w_C)$ denotes the Upstream Hegemon's exogenous technological productivity, decreasing in the input price w_C (the formal decomposition $\theta = \bar{w}/(w_C - m)$, with $\bar{w}$ and m as primitives, is introduced in §3.4 below).⁵ It scales the Security Utility for a microfounded reason: under the probabilistic kill-switch interpretation (Appendix B.2), the potential damage that alignment averts is itself increasing in the Upstream Hegemon's productivity, so security benefits λx are scaled by θ . Economically, as the bottleneck tightens—the Upstream Hegemon's critical input becomes cheaper—the strategic salience of alignment rises proportionally.

Implementation problem. The Principal cannot choose x directly; it must induce x through the schedule $R(x)$. To induce a target alignment x , the US must satisfy two constraints:

1. *Incentive Compatibility (IC)*: $R'(x) = C'(x)$.
2. *Participation Constraint (PC)*: $R(x) \geq C(x) + V_{out}$, where V_{out} is the Swing State's outside option defined in §3.2.

Since the US minimizes the transfer $R(x)$, the PC binds at the optimum, giving the *Implementation Cost Function*

$$R(x) = C(x; w_C) + V_{out} = \frac{(1-x)w_C + xw_A}{A(x)} + V_{out}, \quad (6)$$

where $A(x)$ is the CES aggregator from (2). Following the standard Principal-Agent approach with observable actions (Grossman and Hart, 1983), we treat this cost as the effective price of alignment. When the IC condition binds with indifference at the target x^* , we adopt the standard forcing-contract convention: Hegemon U can impose an arbitrarily small off-path penalty that selects x^* as the Agent's unique best response.

⁵ θ is primitive in Stage 2: Hegemon C chooses m in Stage 1 taking θ as given, so $\theta(w_C)$ is well-defined once the equilibrium w_C is in hand.

Substituting the implementation cost into the objective function and letting $J(x) \equiv V_{US}(x) - C(x; w_C)$ denote the *Joint Economic Surplus*, the US problem reduces to

$$\max_{x, \lambda} \quad \mathcal{W}_{reduced} = \lambda x \theta(w_C) + (1 - \lambda)[J(x) - V_{out}] - \frac{\gamma}{2} \lambda^2. \quad (7)$$

Optimal policy and doctrine. We characterize the equilibrium by maximizing the reduced objective function (7) with respect to the alignment target x and the security doctrine λ . The solution is determined by the intersection of two reaction functions.

First, consider the optimal induced alignment $x^*(\lambda)$. Taking the security weight λ as given, the US chooses the target x^* to equate the marginal political gain with the marginal economic loss:

$$\frac{\partial \mathcal{W}}{\partial x} = \lambda \theta(w_C) + (1 - \lambda) J'(x^*) = 0 \quad \implies \quad J'(x^*) = -\frac{\lambda \theta(w_C)}{1 - \lambda} \quad (8)$$

This condition implies that as the security doctrine becomes more hawkish ($\lambda \rightarrow 1$), the US pushes the target x^* further into the economically inefficient region where the marginal loss of joint surplus $J'(x)$ is high. The threat multiplier $\theta(w_C)$ amplifies this effect: a more productive Upstream Hegemon (higher θ) shifts x^* upward at every level of λ .

Second, consider the endogenous regime choice $\lambda^*(x)$. Differentiating with respect to λ , we obtain the optimal doctrine rule:

$$\frac{\partial \mathcal{W}}{\partial \lambda} = x \theta(w_C) - [J(x) - V_{out}] - \gamma \lambda = 0 \quad \implies \quad \lambda^*(x) = \frac{x \theta(w_C) - [J(x) - V_{out}]}{\gamma} \quad (9)$$

Equation (9) defines the *Strategic Gap*: the difference between the threat-weighted Security Benefit ($x \theta(w_C)$) and the Net Economic Surplus ($J(x) - V_{out}$). The optimal doctrine λ^* is proportional to this gap. A rising Upstream Hegemon (lower w_C , higher θ) widens the gap through two channels. First, it raises the perceived threat intensity θ . Second, it compresses the Net Economic Surplus by raising the implementation cost $R(x)$ of re-routing supply

chains.

Proposition 1 (The Regime Switch Mechanism). *The equilibrium is the fixed point (λ^*, x^*) of these two reaction functions. The nature of the equilibrium depends on the magnitude of the Joint Surplus $J(x)$:*

- *Generous Empire: If $J(x)$ is large (trade is profitable), the Strategic Gap is small or negative. Consequently, $\lambda^* \rightarrow 0$, and the US relies on market integration.*
- *Coercive Hegemon: If $J(x)$ collapses (due to rising costs $C(x)$ or falling value V_{US}), the Strategic Gap widens. The optimal λ^* rises continuously from the corner $\lambda^* = 0$, with a kink at the threshold $\bar{\theta}$ where the corner constraint becomes slack; above the threshold, λ^* increases smoothly toward coercion.*

Existence, uniqueness, and local stability. The equilibrium (λ^*, x^*) is well-defined. Appendix B.1 verifies that the Stage-2 objective is strictly concave in (x, λ) whenever γ is sufficiently large—i.e., when the institutional cost of securitization dominates the cross-effect. This concavity yields a unique interior fixed point that is locally stable under best-response dynamics. Economically, the “ γ sufficiently large” condition means that doctrine and alignment do not amplify each other into tipping points; when γ is too small, multiplicity arises and the economy can jump discontinuously between peaceful and coercive equilibria.

3.4 Stage 1 (Upstream Hegemon): Pricing Strategy

In the first stage, Hegemon C sets the terms of trade for the critical input. To distinguish between technological fundamentals and strategic intent, we decompose the input price into production costs and industrial policy.

Technology and industrial policy. Let w_C be the final export price of the input. We decompose this price into an exogenous technology component and an endogenous policy component:

$$w_C = \underbrace{\frac{\bar{w}}{\theta}}_{\text{Marginal Cost}} + \underbrace{m}_{\text{Strategic Markup}} \quad (10)$$

where $\bar{w}$ denotes the baseline factor price. The parameter $\theta > 0$ represents the *Exogenous Technological Productivity*; an increase in θ captures the structural phenomenon of “China’s Rise” or technological catch-up. The variable m represents the *Endogenous Industrial Policy*, which captures the state’s strategic intent. A positive value ($m > 0$) indicates a standard profit-maximizing monopolistic markup, whereas a negative value ($m < 0$) indicates a strategic state subsidy (or dumping) designed to capture market share.

Hegemon C’s optimization. Normalizing final output $Y = 1$, Hegemon C chooses the markup m (and thus the price w_C) to maximize a weighted sum of commercial profit and geopolitical influence:

$$\max_m \mathcal{W}_C = \underbrace{m \cdot Q_C(x(w_C))}_{\text{Net Commercial Return}} + \underbrace{\phi(1 - x(w_C))}_{\text{Geopolitical Influence}} \quad (11)$$

where $Q_C(x) = \frac{1-x}{A(x)}$ is the derived demand for Chinese inputs. The objective highlights the trade-off: lowering m (subsidies) incurs a fiscal cost but increases the strategic value $\phi(1 - x)$ by binding the Swing State to the Red Supply Chain.

Geopolitical elasticity. Hegemon C faces a derived demand curve $Q_C(x(w_C))$ whose elasticity ε_D is composite. It depends on (i) the technological substitution σ and (ii) the strategic reaction of the US. This creates a policy feedback loop: if Hegemon C lowers w_C too aggressively, it can trigger a coercive regime switch ($\lambda^* \uparrow$) that forces x upward, contracting Chinese export volume Q_C despite the lower price.

The optimal price w_C^* is characterized by a modified Monopoly Pricing Rule:

$$\frac{w_C^* - (\bar{w}/\theta)}{w_C^*} = \underbrace{-\frac{1}{\varepsilon_D}}_{\text{Monopoly Markup}} - \underbrace{\frac{\Phi_{strat}}{w_C^*}}_{\text{Geopolitical Discount}} \quad (12)$$

where $\Phi_{strat} > 0$ represents the marginal strategic value of market share. Because influence matters ($\phi > 0$), Hegemon C optimally sets a price *lower* than the standard profit-maximizing monopoly level, effectively engaging in predatory pricing to secure the vertical supply chain.

3.5 General Equilibrium: The Transmission of the Shock

Having characterized the optimal strategies of the Upstream Hegemon (Pricing), the Downstream Hegemon (Doctrine), and the Swing State (Alignment), we now close the model. We analyze how an exogenous increase in Chinese productivity ($\theta \uparrow$) reverberates through the vertical supply chain and reshapes the global security architecture.

Proposition 2 (The Paradox of Competitiveness). *An exogenous rise in Upstream Productivity ($\theta \uparrow$, “China’s Rise”) necessarily triggers an endogenous rise in the Downstream Security Doctrine ($\lambda^* \uparrow$, “US Hawkishness”). The resulting equilibrium is coercive.*

Proof sketch. The shock transmits through four links. First, higher θ lowers Hegemon C’s marginal cost $\bar{w}/\theta$. Second, Hegemon C re-optimizes: since both commercial profit and the geopolitical influence motive ($\phi > 0$) favor capturing market share, the optimal equilibrium price falls, $dw_C^*/d\theta < 0$. Third, the cost-of-decoupling gap $\Delta C(x) = C(x) - C(x_a)$ widens because $\partial C/\partial w_C = (1-x)/A(x)$ is larger at low x than at high x , so a fall in w_C reduces the Red-Chain-intensive base more than the Blue-Chain-intensive tail (Lemma 1). Fourth, the Strategic Gap in (9) widens and λ^* rises: $d\lambda^*/d\theta = (\partial\lambda^*/\partial w_C)(dw_C^*/d\theta) > 0$, both factors negative. Full derivation in Appendix B.5. $\square$

Corollary: the inevitability of coercion. The transition to coercive trade policy is not necessarily driven by ideological shifts or “unfair” industrial policy (m). Even if Hegemon C

behaves as a benign market actor (holding m constant), a pure productivity shock is sufficient to trigger the regime switch.

We term this dynamic the *Tragedy of Efficiency*: the more efficient China becomes, the harder it is for the US to compete through positive inducements such as subsidies or market access. Once the required subsidy exceeds US fiscal or political capacity, the optimal strategy flips from integration (low λ , high R) to coercion (high λ , negative R). The US then uses tariffs not to protect its own industries, but to artificially destroy the Rival's efficiency advantage, restoring the relative attractiveness of the Blue Chain.

3.6 Welfare: The Tragedy of Efficiency

A claim of tragedy requires a welfare statement. Let $\mathcal{S}(x; \theta)$ denote the joint economic surplus of the three players: Hegemon U's gross value plus Hegemon C's commercial profit, minus the Swing State's Decoupling Cost. The formal decomposition is Proposition 7 in Appendix B.5. The integrated planner's *first-best alignment* $x^{FB}(\theta)$ maximizes $\mathcal{S}$ while ignoring the security externality, and is *declining* in θ : a more efficient Upstream Rival raises the opportunity cost of decoupling. The equilibrium $x^*(\theta)$ moves in the opposite direction (Proposition 2). Define the *efficiency wedge*:

$$\Delta(\theta) \equiv \mathcal{S}(x^{FB}(\theta); \theta) - \mathcal{S}(x^*(\theta); \theta) \geq 0. \quad (13)$$

Proposition 3 (The Tragedy of Efficiency). *Under Assumption 2 and in the coercive regime ($\lambda^* > 0$), the efficiency wedge $\Delta(\theta)$ is strictly increasing in Upstream productivity θ . That is, improvements in Upstream technology generate larger global welfare losses relative to the integrated first-best.*

Proof sketch. In the coercive regime, $x^*(\theta)$ rises with θ while $x^{FB}(\theta)$ falls; the gap widens monotonically, and strict concavity of $\mathcal{S}$ at x^{FB} delivers the result. The full welfare decomposition—including the curvature and second-order-markup conditions under which

the commercial component is strictly decreasing in θ —is in Appendix B.5. $\square$

Interpretation. Proposition 3 is the formal sense in which “Tragedy of Efficiency” is literal. Under horizontal rivalry, a pure productivity gain in the Upstream Rival would be unambiguously welfare-improving. Here, instead, it triggers an equilibrium response in Downstream doctrine that destroys commercial surplus faster than the gain accrues. The Swing State bears the bulk of the loss, because the coercive corner truncates its choice set.

Why verticality, not horizontality, leads to coercion. Under horizontal rivalry—whether competing final goods (subsidy war) or competing sourcing (bidding war)—the cost of retaining the Swing State’s loyalty is linear in the Rival’s efficiency gain, and the security parameter λ stays at zero. Under the vertical bottleneck (Assumption 2), the Decoupling Cost $C(x; w_C)$ is strictly convex and asymptotic on the decoupling range (Lemma 1). The subsidy required to induce alignment $x \rightarrow 1$ therefore explodes as $w_C \downarrow$: the Hegemon cannot buy loyalty and must instead punish disloyalty. Appendix B.8 develops the three-case comparison in detail.

3.7 Leakage Asymmetry Across Sectors

The regime transition in Proposition 1 characterizes the Hegemon’s choice in a single sector. With multiple sectors of differing σ_k (critical, strategic, commodity), the intensity of substitution following coercion differs sharply by bucket. The same security shock $d\lambda$ that fully displaces Chinese supply in commodity sectors displaces it only partially in critical sectors, because the Swing State cannot absorb the displacement without re-importing Chinese intermediates. Through this indirect leakage channel, Chinese value-added reaches the Hegemon via swing-state re-exports rather than direct shipments.

Scope: sectoral separability. The multi-sector analysis that follows treats sectors as structurally decoupled: (i) each sector k has its own CES input mix between Chinese and

non-Chinese intermediates, with no cross-sector substitution; (ii) the Hegemon sets its security intensity λ_k sector by sector, with no joint fiscal or administrative constraint linking λ_c and λ_m ; and (iii) tariffs τ_j and the policy schedule $R_k(x_k)$ operate within a sector, with no cross-sector general-equilibrium price spillovers. This sectoral separability is the minimum assumption needed for a composition margin that is distinct from the flow margin. Relaxing (ii)—adding a shared coercion budget—would push the Hegemon to concentrate λ where bottlenecks are severest, which strengthens the leakage asymmetry rather than overturning it. Relaxing (i) or (iii) opens a multi-sector general-equilibrium problem that is outside the scope of this paper; we note it as an extension and treat the segregated benchmark as the identification target here.

Proposition 4 (Leakage Asymmetry (Flow Margin)). *Let two sectors $k \in \{c, m\}$ differ only in their CES substitution elasticity, with $\sigma_c < 1 < \sigma_m$ (critical vs. commodity). Suppose an exogenous rise in λ induces displacement of Chinese supply. The equilibrium reduction in U 's direct imports from China is partially offset by Swing-State re-imports from China in the critical sector ($k = c$), whereas in the commodity sector ($k = m$) substitution is near-complete. Formally, letting $\Delta M_k^{C \rightarrow U}(\lambda)$ denote the equilibrium change in U 's direct China imports and $\Delta M_k^{C \rightarrow S}(\lambda)$ the change in Swing-State China imports,*

$$\left. \frac{\partial \Delta M_k^{C \rightarrow S} / \partial \lambda}{\partial |\Delta M_k^{C \rightarrow U}| / \partial \lambda} \right|_{k=c} > \left. \frac{\partial \Delta M_k^{C \rightarrow S} / \partial \lambda}{\partial |\Delta M_k^{C \rightarrow U}| / \partial \lambda} \right|_{k=m}.$$

That is, the leakage ratio is strictly positive in the bottlenecked sector ($k = c$) and weakly negative in the commodity sector ($k = m$); the sign flips across the substitution threshold $\sigma_k = 1$. The claim is a directional statement about this sign structure, not a magnitude comparison of two positive quantities.

Proof (Sketch). The Swing State's cost-minimization problem is CES with substitution elasticity σ_k between Chinese and non-Chinese intermediates. A rise in λ raises the effective shadow cost of the Chinese input only at the U–S interface (via the menu schedule $R(x)$),

not at the C–S interface. The Swing State’s optimal mix therefore shifts more sluggishly in low- σ sectors, since substitution toward non-Chinese intermediates is bounded above by $\sigma_k \cdot (d \ln \text{relative price})$. For $\sigma_k < 1$, residual demand for Chinese intermediates is inelastic; the Swing State continues sourcing them and re-exports the embodied content to the U, which is the leakage channel. For $\sigma_k > 1$, the Swing State can switch suppliers cheaply, so the leakage ratio shrinks toward zero. The full derivation is in Appendix B.6. $\square$

Proposition 5 (Leakage Asymmetry (Composition Margin)). *Let $\text{FVA}_{s,j}^C$ denote the Chinese value-added share in country s ’s stationary-equilibrium gross exports to Hegemon U in industry j .⁶ Let $\tau_j \geq 0$ denote the Hegemon’s tariff on direct Chinese exports in j , and let $s \in \{S, N\}$ index Swing versus Non-swing origins. Assume:*

- (A1) *Complementarity. $\sigma_c < 1$ in the critical production structure (same CES primitive as Proposition 4).*
- (A2) *Baseline dependence. The Swing State’s baseline Chinese-intermediate share strictly exceeds the Non-swing origin’s: $\alpha_S > \alpha_N$.*
- (A3) *Empirically relevant range. Baseline Chinese-content shares in swing-state exports to U satisfy $s_S^C, s_N^C \in (0, \frac{1}{2}]$.*
- (A4) *Ordered pass-through. Tariff pass-through to effective Chinese-input price at the C–S interface is weakly larger for the Swing State than for the Non-swing origin: $\eta_S \psi'_S \geq \eta_N \psi'_N$, where η_s is the Chinese-intermediate share and ψ'_s the marginal pass-through from τ_j to the Swing’s/Non-swing’s Chinese-input shadow price.*

Then the steady-state FVA-share satisfies the positive cross-partial

$$\left. \frac{\partial^2 \text{FVA}_{s,j}^C}{\partial \tau_j \partial \mathbf{1}\{s = S\}} \right|_{\sigma_c < 1} > 0.$$

⁶ “Stationary equilibrium” here denotes the cross-sectional industry-partner pattern conditional on the vector of primitives $(\tau_j, \alpha_s, \sigma_c, \lambda)$; the model has no explicit transitional dynamics and we invoke stationarity solely to interpret the object identified empirically from cross-section variation.

Equivalently, across industries that differ in τ_j , the tariff-gradient of the embodied-Chinese share is strictly steeper for swing-state exporters than for non-swing exporters.

Remark on (A3)–(A4). (A3) restricts baseline Chinese-content shares to $(0, \frac{1}{2}]$, which matches the empirically relevant range—the swing-state FVA-China share in critical-sector exports to the U is approximately 6% in our OECD ICIO / ADB MRIO sample, well inside $(0, 1/2]$ (Appendix B.7). The restriction is load-bearing: the slope $\partial s^C / \partial \ln p = (1 - \sigma_c) s^C (1 - s^C)$ is increasing in s^C on $(0, 1/2]$ and decreasing on $[1/2, 1)$, so the sign of the cross-partial flips if the baseline share exceeds $1/2$. We therefore state Proposition 5 as a conditional result valid on the empirically relevant range and do not claim it globally. (A4) is a symmetric-pass-through condition: it holds whenever the Swing State’s larger baseline Chinese-intermediate share translates into weakly larger sensitivity of its effective Chinese-input shadow price to τ_j . We state it as an explicit hypothesis rather than derive it from (A2), because the mapping from α_s to $\eta_s \psi'_s$ depends on the Swing State’s *downstream* bargaining structure vis-a-vis U, which is outside the scope of this paper. Under (A2) alone, (A4) is a natural ordering but not a theorem. Proposition 5 is therefore a conditional statement: it identifies the cross-partial under (A1)–(A4), and the sign of the empirical moment we test is the sign predicted by the proposition if and only if (A4) holds in the data. Finally, the cross-partial in Proposition 5 is a group-average object. Reclassifying a single origin between $\{S, N\}$ leaves the group moment unchanged up to $O(1/|S|)$, so the identification is robust to borderline-country manipulation of the swing-state partition—a counter-fragility property our leave-one-out diagnostic confirms empirically.

Proof (Sketch). The Swing State’s cost-minimizing input mix is CES with elasticity $\sigma_c < 1$ between Chinese and non-Chinese intermediates. A higher τ_j raises the shadow price of *direct* Chinese supply at the U–C interface but not at the C–S interface, so the Swing State’s effective Chinese-input price rises by less than the Hegemon’s. With $\sigma_c < 1$, the Swing State cannot cheaply rebalance away from Chinese intermediates, and a larger fraction of the Hegemon’s tariff-induced displacement is absorbed as re-exported Chinese content in j . The

Non-swing origin, with strictly smaller baseline Chinese-intermediate share, absorbs less per unit of τ_j reallocation. Crucially, the claim is about the *cross-partial* (the tariff-gradient is steeper for S), not merely the level (the FVA share is higher for S). This follows because, under $\sigma_c < 1$, the slope of the CES Chinese-input expenditure share with respect to the effective relative price is $\partial s^C / \partial \ln p = (1 - \sigma_c) s^C (1 - s^C)$, which is strictly increasing in the baseline share s^C on the range $(0, \frac{1}{2}]$ specified by (A3); equivalently, the share function is concave in the effective relative price on this range. At the Swing State’s higher baseline Chinese-input share (A2), the same $d\tau_j$ therefore moves the Swing State along a steeper segment of the cost-minimization surface. It therefore displaces a larger quantum of Chinese value-added *back through* re-exported intermediates than the Non-swing origin, which operates at a flatter segment. The cross-partial is strictly positive whenever $\sigma_c < 1$ and the Swing State’s baseline Chinese-intermediate share strictly exceeds the Non-swing origin’s. The full derivation is in Appendix B.7. $\square$

Empirical correlates. Propositions 4 and 5 generate two complementary moments tested in Section 4: a flow-margin event study exploiting the 2018 tariff shock as an identified λ -increment, and a cross-sectional triple-difference that identifies the steady-state cross-partial $\partial^2 \text{FVA}_{s,j}^C / \partial \tau_j \partial \mathbf{1}\{s = S\}$ without requiring an isolated shock. The cross-sectional moment is the primary one: it does not depend on the sharpness of the 2018 break, and the cross-partial is robust to borderline-country reclassification up to $O(1/|S|)$.

3.8 Robustness and Extensions

Reconciling the kink with smooth aggregate dynamics. Proposition 1 predicts a continuous-but-nondifferentiable kink in $\lambda^*(\theta)$ at $\bar{\theta}$, while the empirical dynamics (e.g., Figure 3) appear smooth. Three non-exclusive reconciliations—heterogeneous sectors crossing the threshold sequentially, Bayesian uncertainty about σ smoothing the expected λ^* , and institutional adjustment costs on coercive capacity—are developed in Appendix B.9.

Three baseline extensions. Three extensions relax baseline modelling choices without altering the core results.

First, Nash-bargaining power β for the Swing State *amplifies* coercion: more powerful swing states trigger the regime switch sooner (Proposition 6; Appendix B.3). This rationalizes the US pattern of combining subsidies (CHIPS Act) with coercive measures for allies with high β (Korea, Taiwan, the Netherlands).

Second, endogenizing vertical position via a specialization pre-stage replicates the baseline structure when $\theta_C > \theta_U$ and $E_U > E_C$. The vertical asymmetry is therefore a Ricardian outcome, not a modelling choice (Appendix B.4).

Third, a probabilistic kill-switch microfoundation reinterprets λ as the expected damage averted by alignment, connecting to the security-dilemma logic: $d\lambda^*/d\theta > 0$ means Upstream efficiency endogenously raises the Hegemon’s perceived probability of weaponization (Appendix B.2).

4 Empirical Tests of the Leakage Mechanism

Reading Fact 4 through the model. The coercion variable λ^* maps onto Fact 4’s two Entity-List series differently. The China-bilateral line (which kinks at 2018) is the direct Regime-Switch signature: $\lambda^*(\theta)$ crosses $\bar{\theta}$ once Section-301 raises θ past the Generous-Empire corner. The third-country line (constant log-slope throughout the sample) is not λ^* itself but the administrative infrastructure that enforces it—the primitives γ and $\bar{T}$ in the model. Its steady growth is therefore consistent with the model, not evidence against the 2018 kink.

With the model of Section 3 and Propositions 4 and 5 in hand, we subject the leakage mechanism to two complementary regression-based tests.

Test 1 (the HS6-panel event study) targets the flow-margin prediction of Proposition 4. If Section-301 triggered a bilateral China–critical decoupling, the post-2017 coefficient path should turn sharply negative. If swing-state intermediation had already absorbed the margin

before the shock, the path should instead be flat against the 2017 baseline, with a pre-existing negative pre-period.

Test 2 identifies the cross-partial of Proposition 5 using a continuous tariff- $\times$ -swing triple-difference on Chinese value-added shares in swing-state $\rightarrow$ US gross exports. Under $\sigma_c < 1$, the tariff-gradient of Chinese-VA content in swing-state exports is steeper than in non-swing exports. This is a *steady-state* object that does not require isolating a 2018 shock. A third dose-response regression on the Entity-List coercion stock (Spec 2) tests the paradox mechanism of Proposition 2 as an auxiliary margin.

Headline, previewed. Test 1 returns a null—not a refutation of leakage but its signature. The pre-2018 critical-commodity gap was already negative and *closes* toward zero after 2018, consistent with swing-state intermediation having absorbed the bilateral margin before the tariff shock. Robustness to group-specific trends, Rambachan-Roth sensitivity, and Poisson zero-retention leaves this reading intact. Test 2 then identifies the leakage cross-partial at $\hat{\beta}_{\tau \times \text{swing}}^{\text{post}} = +0.102$ ($p < 0.001$). A 25 pp Section-301 tariff increase on a swing-state-produced good is associated with a further +2.5 pp rise in its Chinese-VA content—a steady-state fingerprint of the $\sigma_c < 1$ production structure. Placebo-year diagnostics reframe this as cross-sectional rather than quasi-experimental identification (the gap pre-existed 2018), which is the correct reading of a steady-state proposition.

We construct a balanced panel of bilateral US imports at the HS6 level over 2000–2024, assigning each code to one of three buckets—*critical* (rare earths, advanced semiconductors, semiconductor-manufacturing equipment), *strategic* (mature-node chips, batteries, EV components), and *commodity* (textiles, basic steel, basic chemicals). Appendix C details the HS6 concordance, the UN Comtrade pull (premium API, 63 targeted HS6 codes, 18 reporters), and the BIS Entity List panel built from the OpenSanctions Consolidated Screening List.

Mapping specifications to propositions. Specs 1 and 3 test coefficients implied by the leakage asymmetry of Proposition 4; Spec 2 (the “paradox regression”) is a dose-response

test of the sector-asymmetric coercion effect. All three specifications isolate different pieces of the same underlying production structure, differing in how they collapse the data over time.

- **Spec 1 (triple-difference, single Post dummy).** Tests the *average* post-2018 differential. A naive reading of Proposition 4: $\beta_{\text{CHN} \times \text{Crit} \times \text{Post}} < 0$ —US imports from China fall more in critical than commodity sectors once λ rises. This is the “tariffs caused decoupling” sign that a casual reader would expect.
- **Spec 2 (paradox regression).** Tests the *dose-response* of the sector-asymmetric effect to the Entity-List coercion stock $\text{EL}_{t-1}^{\text{CHN}}$. Under Proposition 4, the interaction with the critical-sector dummy should load negatively: each additional unit of coercion should depress Chinese critical imports more than commodity imports.
- **Spec 3 (event study).** Tests the *dynamic* sector-asymmetric path year by year. Under Proposition 4, with swing-state intermediation already operating pre-2018, the $(\text{CHN} \times \text{Critical})$ coefficient path should show two features. First, it should be *negative before 2018*, because the gap is a steady-state feature of the global production structure, not a 2018 policy break. Second, it should be *flat or rising from below toward zero thereafter*, because additional coercion does not re-open a gap that leakage has already normalized.

Spec 1 and Spec 3 are not equivalent: Spec 1 turns on whether the post period is below zero, while Spec 3 turns on whether the pre period was below zero and the post period converged. The two can point in different directions, and in our HS6 panel they do—Spec 1 is mildly positive while Spec 3 recovers the closure-of-pre-existing-gap pattern Proposition 4 implies when leakage is a steady-state feature rather than a post-shock adjustment. Spec 2’s sector-asymmetric loading, sharp in the 3-sector aggregate, dissolves at HS6 resolution.

4.1 Test 1: Flow-Margin Event Study (Proposition 4)

Using this panel, we estimate an event-study specification for US imports from China in the critical bucket relative to a (swing-state $\times$ non-critical) base, with 2017 as the omitted pre-treatment year. Identification is at the HS6 $\times$ partner $\times$ year level ($N \approx 17,600$). Fixed effects absorb partner-HS6, HS6-year, and partner-year variation, and standard errors are clustered at HS6 (~ 60 clusters):

$$\ln M_{ihst} = \alpha_{ih} + \gamma_{ht} + \delta_{it} + \sum_{\tau \neq 2017} \beta_{\tau} \cdot \mathbf{1}\{t = \tau\} \cdot \text{CHN}_i \cdot \text{Critical}_h + \varepsilon_{ihst},$$

where i indexes exporter (China + 17 candidate swing-state partners; the focal 3-state swing group {VNM, MEX, MYS} used in Test 2 is a subset), h indexes HS6 code, and t indexes years. HS6 resolution (relative to a 3-sector aggregate) raises sample size by a factor of 13 and permits cluster-robust inference on a genuine number of clusters.

The purpose of the event study is not to detect a 2018 break but to ask whether one *should* have existed. If Section-301 caused China-critical decoupling, the post-2017 β_{τ} path should turn sharply negative. Under the leakage reading of Proposition 4, the pre-2018 path was already negative because swing-state intermediation had absorbed the bilateral margin. The post-2018 path should then be flat against the 2017 baseline rather than deepen. The data match the latter. Pre-2018 coefficients sit at -0.3 to -0.8 log points, with 2012–2016 individually significant at 1–5%. Every post-2017 coefficient is statistically indistinguishable from the 2017 baseline. The joint pre-trend test $F_{5,61} = 3.69$ ($p = 0.006$) rejects a flat pre-period, which is exactly the pre-existing gap the leakage story predicts. See Appendix Figure 8 for the full coefficient path and Appendix Table 2 for the numerical values.

The headline single-coefficient triple-difference $\beta_{\text{CHN} \times \text{Critical} \times \text{Post}} = +0.283$ (SE 0.245, $p = 0.25$) is statistically insignificant with a positive sign. This mirrors the event study: the post-2018 coefficients cluster around zero while pre-2018 coefficients are negative, so the collapsed Post contrast loads weakly positive. The HS6 panel does not support a *widening* of the

China–Critical gap after 2018; it supports the gap *closing*.

Trend-robustness. Column 2 of Table 2 adds group-specific linear trends interacting ($\text{CHN} \times \text{Critical}$) and ($\text{Swing} \times \text{Sector}$) with time. The absorbed China–Critical trend is -0.037 log points per year (SE 0.028, not significant); the residual Post interaction is $+0.684$ ($p = 0.013$). Even with the trend taken out, there is no evidence of a *negative* post-2018 residual effect on China–critical imports. The interpretation most consistent with this evidence is that by 2018 the China–critical gap had already closed via swing-state intermediation, and subsequent coercion neither re-opened it nor slowed the convergence. This is consistent with the *leakage* mechanism of Proposition 4—swing states absorbed Chinese intermediate capacity into the US supply chain—and inconsistent with a simple “tariffs caused decoupling” reading.

Pre-trend sensitivity (honest DiD). Because the pre-trend F -test rejects strict parallel trends, we report robust identified sets for the average post-treatment effect using the relative-magnitudes partial-identification procedure of Rambachan and Roth (2023), which bounds the post-period deviation as a multiple M of the worst pre-period deviation. Two features matter (Appendix Figure 9; numerical bounds in Appendix C.4). First, *zero lies inside the identified set at every $M \leq 2$* , so the hypothesis that the post-2018 China–Critical gap equals the 2017 baseline is not rejected even under very weak parallel-trends assumptions. Second, the *upper* bound is positive throughout, ruling out any persistent post-2018 *widening* of the gap. The leakage reading—closure rather than widening—is thus not dependent on the parallel-trends assumption that the event-study literally rejects.

Zero-retention (PPML). The log specification drops cells with nonpositive trade, a selection disproportionately concentrated in $\text{China} \times \text{Critical} \times \text{post-2020}$ —exactly the cells where decoupling would generate new zeros. We therefore re-estimate on a balanced ($\text{HS6} \times \text{partner} \times \text{year}$) panel with missing cells set to zero, using the Poisson pseudo-

maximum-likelihood estimator of Santos Silva and Tenreyro (2006) via `ppmlhdfc` (Correia et al., 2020). On $N = 22,305$ observations (a 26% increase over the log sample), the baseline triple-difference coefficient is +0.289 (SE 0.309) and the trend-robust variant is +0.531 ($p = 0.050$). Both are close to their log-sample counterparts (+0.283 and +0.684) and, if anything, slightly weaker. The PPML event study (Appendix C, Table 4) reproduces the key qualitative pattern: negative pre-2018 coefficients (the largest, $\tau_{2015} = -0.400$, is individually significant) and a post-2018 band clustered around zero with no persistent negative drift. The zero-retention critique therefore does not rescue a “tariffs caused decoupling” reading; the gap-closing pattern is not an artifact of selecting on positive trade.

4.2 Test 2: Steady-State Cross-Partial on Chinese Value-Added (Proposition 5)

The gross-trade test above ruled out a persistent post-2018 widening of the US–China gap but does not directly demonstrate Chinese content flowing through swing states. The value-added decomposition does.

Direct evidence from value-added decomposition. The triple-difference in gross trade is an *indirect* test of Proposition 4. A closing US–China bilateral gap in critical-sector imports can reflect leakage through swing-state intermediation, but only if swing-state exports to the US embed more Chinese value-added in critical sectors than in commodity sectors. We test this directly. We use two separately maintained inter-country input–output tables: the OECD ICIO EXTENDED release (ISIC Rev. 4, 50 industries, 85 economies, 1995–2022) and the ADB MRIO 72-economy release (35 industries, 2017–2024). For each swing state s , industry j , and year t , we compute the share of gross $s \rightarrow$ US exports accounted for by Chinese value-added:

$$\text{FVA}_{s,j,t}^{\text{CHN, share}} = \sum_{r \in \text{CHN}} v_r \cdot L_{r,(s,j)},$$

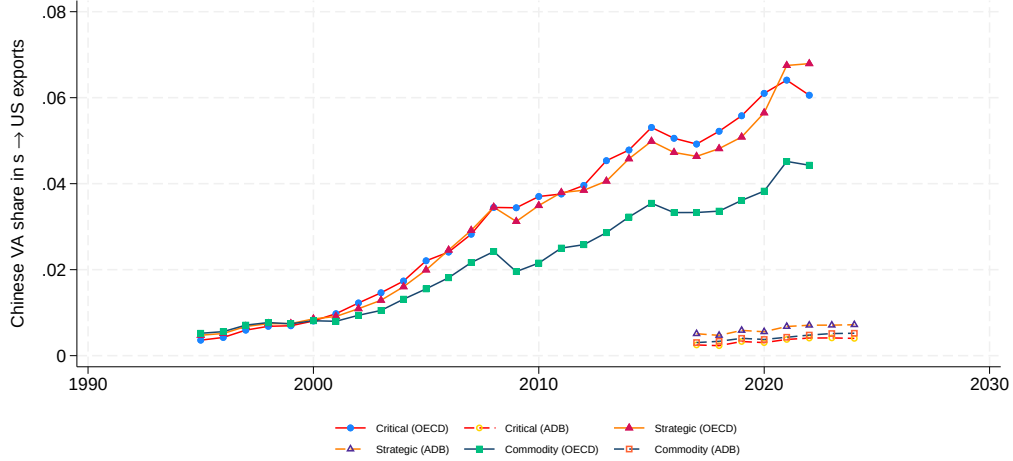


Figure 5: **Chinese value-added share in swing-state $\rightarrow$ US exports, by sector bucket.** Cross-swing-state mean of $FVA_{s,j,t}^{CHN}/E_{s,j,t}$, computed from OECD ICIO EXTENDED (solid, 1995–2022) and ADB MRIO 72-economy release (dashed, 2017–2024). ADB levels are $\approx 20\times$ lower because its 35-industry taxonomy aggregates semiconductors with household appliances in a single “electrical and optical” bucket, diluting the critical share. Levels and trends discussed in the text; full regression in Appendix Table 5.

where v_r is value-added per unit of gross output in Chinese source sector r and $L = (I - A)^{-1}$ is the global Leontief inverse. This share is *multiplied by* gross bilateral exports $E_{s \rightarrow US,j,t}$ to recover the dollar value of Chinese value-added embedded in those exports, which is what we use in the dollar-volume rescaling below. Figure 5 plots the across-swing-state mean by sector bucket and data source.

The level gap is the clearest evidence. In the OECD panel the critical-sector share rises from 0.4% in 1995 to a flat plateau of $\approx 6.1\%$ over 2020–2022, while the commodity share rises from 0.5% to 4.4% over the same period. The critical–commodity gap opens in the mid-2000s and reaches ≈ 1.8 percentage points by 2015–2018. It then *widens* to ≈ 2.3 percentage points by 2020: Chinese value-added embedded in swing-state critical-sector exports to the US kept rising after the 2018 tariffs, even as gross US imports from China fell.⁷

⁷Scaled against 2022 swing-state $\rightarrow$ US gross exports, the fitted shares imply roughly \$22 billion of Chinese value-added embedded in the critical bucket and \$23 billion in the commodity bucket. A back-of-envelope figure for the post-2018 descriptive-plot share acceleration (≈ 0.9 percentage points in critical sectors) is approximately \$3 billion per year; this is a descriptive statistic, not a causal counterfactual, because the placebo-year regressions reported below show that approximately 47–87% of the post-2018 cross-sectional gap was already present before the 2018 tariffs (the earliest 2010 placebo recovers 87% of the 2018 magnitude).

In a stacked regression on $N = 11,899$ (source- $\times$ -industry- $\times$ -dataset, $\times$ year) observations with two-way fixed effects (Appendix Table 5), the post-2018 interaction on the critical bucket is $+0.009$ ($p < 0.05$ pooled; $+0.012$, $p < 0.05$ OECD only).

Disaggregating along Section-301 exposure, the post-2018 acceleration concentrates on *electronics*, the Section-301 flagship target (coded C26 in ISIC Rev. 4, D26 in the EXTENDED letter-prefix release): $\hat{\beta}_{\text{elec}}^{\text{post}} = +0.030$ ($p < 0.01$). The non-electronics critical sub-bucket—metal ores, coded B07 (D07T08 in the EXTENDED release)—moves in the opposite direction.

The sharpest specification is a continuous triple-difference. It absorbs source- $\times$ -year and industry- $\times$ -year shocks, and identifies off the interaction of (published) Section-301 tariff rates with swing-state identity and post-2018 timing (Appendix C.6). We estimate $\hat{\beta}_{\text{tar} \times \text{swing}}^{\text{post}} = +0.102$ ($p < 0.001$): a 25 pp Section-301 tariff increase on a swing-state-produced good is associated with a further $+2.5$ pp rise in its Chinese-VA content *beyond* what industry-year and source-year effects absorb.

The continuous specification is the primary test of Proposition 5, which predicts a cross-partial $\partial^2 \text{FVA}_{s,j}^C / \partial \tau_j \partial \mathbf{1}\{s = S\}$ continuous in τ_j —not a discrete electronics-vs-metals contrast. Consistent with that primacy, the analogous binary triple-difference (post $\times$ electronics $\times$ swing, with metals as the within-industry control and the 14 non-swing sources as the geographic control) is *not* identified under the same FE structure ($\hat{\beta} = +0.014$, $p = 0.36$). The electronics-vs-metals gap and the swing-vs-non-swing gap are each captured fully by the absorbed effects, and no clean bilateral residual survives.

Theoretical mapping. The continuous triple-difference is the identified moment of Proposition 5 (Steady-State Leakage Composition): under $\sigma_c < 1$, the cross-partial $\partial^2 \text{FVA}_{s,j}^C / \partial \tau_j \partial \mathbf{1}\{s = S\}$ is strictly positive, i.e., the tariff-gradient of the Chinese-VA share in swing-state-to-US gross exports is steeper than in non-swing-state-to-US exports. Our $\hat{\beta}_{\text{tar} \times \text{swing}}^{\text{post}} = +0.102$ is the sample analog of this cross-partial, with source- $\times$ -year and industry- $\times$ -year effects absorbing both the swing-specific aggregate dynamics and

the industry-specific tariff dynamics, leaving the bilateral residual as the target object. Proposition 5 is a steady-state corollary of the same production-structure primitive ($\sigma_c < 1$) that drives the flow-margin leakage of Proposition 4, but unlike that flow margin it does not require isolating a 2018 shock. The binary analogue (post $\times$ electronics $\times$ swing) in contrast does *not* identify Proposition 5, because electronics does not vary continuously in τ_j within-category and the post-interaction with swing-state identity is absorbed by the source-year and industry-year fixed effects once the binary electronics indicator is collinear with their tensor product. The continuous tariff-intensity specification is therefore not a robustness check but the primary test.

Pre-trend caveat (placebo diagnostics). Placebo-year regressions (Appendix Tables 6–7) with fake “post” cuts at 2010, 2013, and 2016 (restricted to year ≤ 2017) reveal that the continuous triple-interaction is *not* identified off a 2018 break. The tariff- $\times$ -swing coefficient is already +0.089 ($p = 0.001$) under a 2010 cut, +0.080 ($p = 0.004$) under a 2013 cut, and +0.048 ($p = 0.04$) under a 2016 cut, against +0.102 ($p < 0.001$) for the real 2018 cut. Pre-treatment placebos thus recover 47–87% of the 2018 magnitude (2010 cut: 87%; 2013 cut: 78%; 2016 cut: 47%). The monotonic decay from 2010→2016 suggests the trend was slowing before 2018 and that the 2018 coefficient represents an acceleration of, not a departure from, a structural pattern. The binary triple-diff is uninformative in *both* the placebo and the real window (all four coefficients statistically indistinguishable from zero).

The continuous triple-diff’s $p < 0.001$ headline clusters on the panel identifier (source $\times$ industry $\times$ dataset) and is conservative to alternative inference structures. Two-way clustering on (source, industry) gives $\hat{\beta} = +0.102$ with $\text{SE} = 0.040$ ($t = 2.57$, $p < 0.01$); here the conservative cluster count is $K = \min(17, 28) = 17$, with 17 source countries (3 swing + 14 non-swing) and 28 ISIC Rev. 4 industries. Both the panel-clustered and two-way-clustered variants reject zero at conventional levels. Leave-one-out of any single swing state leaves the coefficient significantly positive, though the point estimate ranges from +0.058 (dropping Vietnam) to +0.144 (dropping Malaysia).

A companion Honest-DiD exercise applied to the pooled FVA event study (Appendix Table 8) reinforces this reading. The robust 95% identified set for the average post-2018 critical effect has an always-positive upper bound (+0.025 at $M = 0$, rising to +0.066 at $M = 2$). But the lower bound is exactly zero at $M = 0$ and crosses into negative territory for any $M > 0$. So the post-2018 FVA effect does not reject zero even under strict parallel trends. We therefore read the FVA evidence as cross-sectional identification of Proposition 5 (the steady-state cross-partial) rather than as quasi-experimental identification of the Section-301 flow shock; the pre-trend pattern is an expected feature of a steady-state test, not a bug.

4.3 Auxiliary Test: Paradox Dose–Response (Proposition 2)

A companion regression of log US imports from China on the lagged Entity-List China-listings stock EL_{t-1}^{CHN} , interacted with sector bucket, tests the paradox mechanism of Proposition 2 directly. The sector-asymmetric loading is sharp on the 3-sector aggregate panel (-0.217 , $p < 0.001$) but dissolves at HS6 resolution. The post-2018 critical-bucket slope is +0.021 (SE 0.061), and the strategic, commodity, and flow variants are null throughout. We therefore treat the HS6 null as the honest benchmark. The paradox mechanism is not ruled out, but a within-HS6 dose–response test fails to identify it once product-level heterogeneity is absorbed. Test 1 (leakage event study) and Test 2 (FVA cross-partial) remain the primary empirical anchors for the sector-asymmetric story; Appendix Figure 12 plots the underlying sector-aggregate series for reference.

What the regression evidence buys. Taken together, the three regression tests sharpen the motivational facts of Section 2 into identified moments of the model.

First, the “Vertical Trap” (Facts 1 & 2) rationalizes the low-substitution-elasticity primitive ($\sigma_c < 1$) between upstream inputs. The persistence of swing-state reliance on Chinese inputs despite US market pull implies that the efficiency gap (ΔC) from the “Red Supply Chain” is substantial and difficult to replace.

Second, the divergence between direct and indirect decoupling (Fact 3) is sharpened by the cross-partial test. The continuous triple-difference identifies $\hat{\beta}_{\tau \times \text{swing}}^{\text{post}} = +0.102$ ($p < 0.001$), which is the sample analog of the steady-state cross-partial of Proposition 5. The swing-state optimizer therefore allocates more Chinese value-added to higher-tariff sectors precisely because the complementarity makes substitution costly.

Finally, the structural break in the Entity List (Facts 4–6) is consistent with the Hege-mon’s security doctrine (λ) being endogenous to depth of integration. The extra-bilateral perimeter documented in Fact 4 preceded the tariff shock, and the post-2018 bilateral intensification is what the paradox mechanism of Proposition 2 predicts when fiscal alignment becomes unbounded.

The regressions therefore neither confirm nor refute the 2018 Section-301 timing—they identify a *steady-state* production-structure signature that the model formalizes. Having identified that signature empirically, we now visualize the regime switch that generates it.

5 Numerical Illustration and Robustness

To visualize the Regime Switch (Proposition 1) and the Paradox of Competitiveness (Proposition 2) that flows from it, we simulate the model under parameters calibrated to the bottleneck-sector literature. We set $\sigma = 0.35$, intermediate between two anchors: the sector-level materials elasticity of approximately 0.1 estimated by Atalay (2017), and the macro-level range 0.5–0.7 used in the production-network calibrations of Baqaee and Farhi (2024). In addition, Boehm et al. (2019) document a short-run elasticity between imported and domestic inputs that is indistinguishable from Leontief, placing an even lower floor on plausible σ at business-cycle horizons. All three estimates place σ comfortably below unity and satisfy the Bottleneck Condition (Assumption 2). The distribution parameter is set symmetrically at $\alpha = 0.5$. We then simulate “China’s Rise” by exogenously decreasing the upstream input price w_C while holding the alternative price w_A and the US market value V_{US} constant.

Parameter	Value	Source / role
σ (CES elasticity)	0.35	Between Atalay (2017) (≈ 0.1) and Baqaee and Farhi (2024) (0.5–0.7)
α (Chinese-input weight)	0.5	Symmetric baseline
w_A (alternative price)	1	Numerator
V_{out} (Swing outside option)	0	Normalization
$\bar{w}$ (factor-price numeraire)	1	Normalization in (10)
V_0 (US baseline value)	0.84	Calibrated to threshold $\bar{w}_C = 1.6$
β (slope of V_{US})	1.0	Calibrated jointly with V_0
γ (security-policy cost)	2.5	Calibrated for smooth λ^* ; passes local Hessian check
ϕ (China geopolitical weight)	0	Isolates productivity shock channel
w_C range	[0.5, 2.0]	Spans both regimes

Table 1: Parameter configuration for the numerical illustration.

Disclosure: calibration target. σ is external (bottleneck-sector literature). The remaining parameters (V_0, β, γ) are calibrated to a single illustrative target: that the regime-transition threshold land at $\bar{w}_C \approx 1.6$. This value corresponds to the present-day calibrated Chinese productivity at which the Section-301 tariff shock of 2018 activated the coercive machinery (Facts 4–6 of Section 2).

The choice is illustrative, not structural. We fix $\bar{w} = w_A = 1$ and $V_{out} = 0$ as normalizations. We choose $\gamma = 2.5$ as the smallest value that delivers a smooth λ^* with no multiplicity on the w_C grid. Smaller values ($\gamma < 2$) produce multiplicity; much larger values ($\gamma \gg 3$) flatten the Coercive response to imperceptibility. We then jointly solve (V_0, β) to land $\bar{w}_C = 1.6$ with the linear specification $V_{US}(x) = V_0 - \beta x$.

Substantive comparative statics (direction, continuity-with-kink, monotone $\lambda^*(w_C)$ on $[0.5, 1.6]$) are robust to this calibration. They rely on Assumption 2 (under which $\sigma < 1$ is a sufficient CES condition) and on the concavity of the Stage-2 reduced objective, both of which hold for any (V_0, β, γ) in a neighborhood of the chosen values. The numerical threshold level $\bar{w}_C = 1.6$ is merely a visual anchor. What Figure 6 demonstrates is the *qualitative* regime switch, not a structural estimate of when it occurred.

Figure 6 plots the equilibrium response of the Downstream Hegemon. The horizontal axis

represents the Rival's efficiency ($1/w_C$). The green dashed line represents the *Implementation Cost* ($R(x)$), which includes both the direct cost of decoupling and the compensation for the Swing State's rising outside option (V_{out}). The red solid line represents the optimal US Security Doctrine (λ^*).

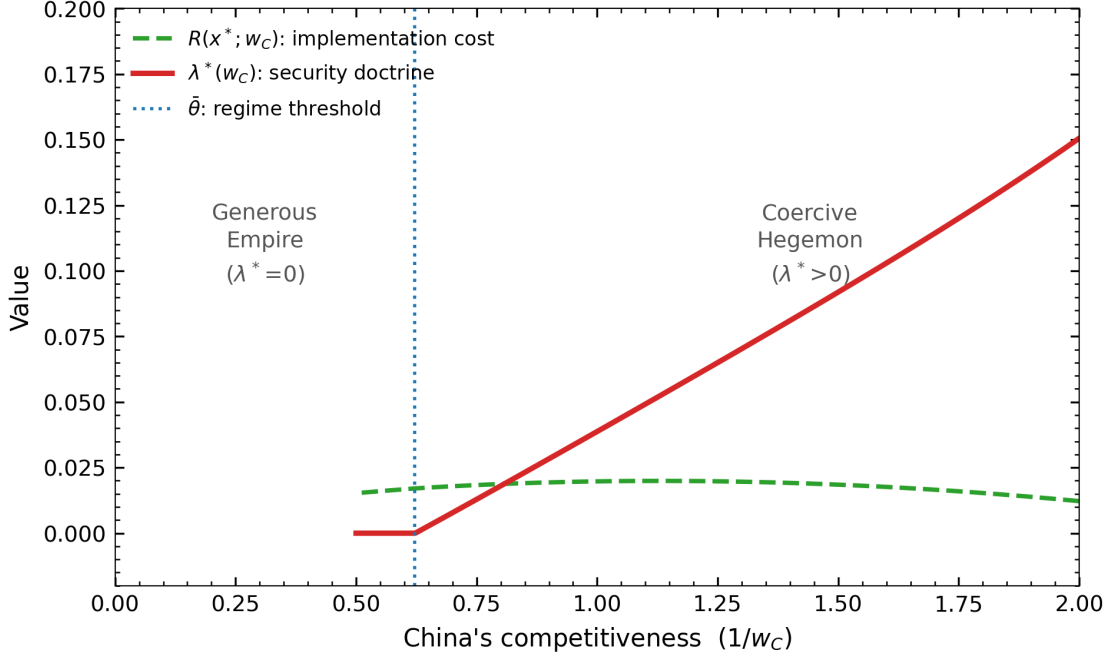


Figure 6: **The Coercive Regime Switch.** As the Rival's efficiency improves (higher $1/w_C$), the subsidy required to buy voluntary alignment (green dashed) rises exponentially. Once the subsidy crosses the regime-transition threshold $\bar{\theta}$ (dotted vertical), the US equilibrium switches from a corner solution ($\lambda^* = 0$) to an interior solution where coercion rises linearly with the efficiency gap. Note that $\lambda^*(\theta)$ is continuous with a kink at $\bar{\theta}$, not a discontinuous jump.

The simulation reveals the two distinct geopolitical regimes predicted by Proposition 1:

Regime 1: The Generous Empire (Corner Solution). On the left side of the figure ($w_C > 1.6$), the cost of decoupling is manageable. The “Strategic Gap” is negative because the economic surplus from trade remains high relative to the security benefit. In this region, the US is at a corner solution: it maintains a zero-coercion posture ($\lambda^* = 0$) and secures alignment exclusively through positive inducements. This corresponds to the post-Cold War era of globalization.

Regime 2: The Coercive Hegemon (Interior Solution). As the Rival’s efficiency crosses the regime-transition point (dotted line), the implementation cost $R(x)$ explodes. Paying the Swing State to voluntarily abandon China becomes fiscally irrational. Consequently, the constraint binds, and the optimal doctrine λ^* detaches from zero. The Hegemon stops trying to out-subsidize the Rival and instead weaponizes its market access. Note that λ^* rises linearly while the cost rises exponentially. This gap reflects the institutional damping parameter γ , which prevents the policy from jumping instantly to maximum coercion.

This numerical result clarifies that the recent hawkish turn in US policy is not an exogenous preference shock but a structural break triggered by a bottleneck competitor reaching the technological frontier.

5.1 Robustness: The Empirical Relevance of $\sigma < 1$

The entire mechanism of our paper hinges on the Bottleneck Assumption $\sigma < 1$. We therefore devote this subsection to two goals. First, we provide an empirical defense of the assumption. Second, we report a robustness check showing how the regime-transition threshold $\bar{\theta}(\sigma)$ (the kink location in $\lambda^*(\theta)$) varies with σ across plausible sectoral values.

5.1.1 Sectoral Evidence on the Elasticity of Substitution

Three bodies of empirical work inform the plausible range of σ for the inputs in which US-China rivalry is concentrated.

First, the Armington/CES trade literature estimates elasticities of substitution across source countries for broad manufactured inputs. Broda and Weinstein (2006) report median elasticities of 3–7 at highly disaggregated HS codes, but the distribution has a long left tail. A substantial share of intermediate goods—particularly chemicals, rare-earth products, and capital equipment—exhibit estimates below unity. Caliendo and Parro (2015) adopt calibrated values around 4 in the aggregate trade block but acknowledge explicitly that this is an upper bound not appropriate for bottleneck intermediates.

Second, the production-network literature has estimated elasticities of substitution across *suppliers of a given input*, which is precisely the object of our model. Boehm et al. (2019) exploit the 2011 Tōhoku earthquake as a natural experiment on supply disruption and estimate that the short-run elasticity of substitution between imported and domestic inputs is statistically indistinguishable from zero (Leontief). Atalay (2017) estimates a sector-level elasticity of substitution across materials inputs of approximately 0.1 using annual manufacturing data, well below the Armington-trade values. Baqaee and Farhi (2024) show that plausible micro-level elasticities of input substitution in production networks lie well below one, with implied macro-level $\sigma \approx 0.5\text{--}0.7$.

What each estimate identifies. The three sources above do not all estimate the same object, and their use as a joint bracket on σ_c deserves one clarifying paragraph.

Boehm et al. (2019) identify the short-run elasticity of substitution between *imported and domestic* inputs at the firm level off a supply-disruption natural experiment. This is the closest match to our σ_c , which is the elasticity between *Chinese and non-Chinese* intermediates.

Atalay (2017) identifies the elasticity of substitution across *materials inputs* at the sector level from annual manufacturing data. This is not origin-differentiated, but it shares the key property of being measured at the production-structure level (below any Armington trade block). It furnishes a bound on how elastic input substitution can plausibly be in the bottleneck sectors the policy phenomenon targets.

Baqaee and Farhi (2024) identify the macro-implied *across-input* elasticity in a calibrated production network. This is not a direct estimate of the across-origin elasticity, but it bounds the aggregate network-implied substitutability.

All three identify different, though closely related, elasticities. All three place the relevant object well below unity for the sectors that drive our mechanism. For our purposes the tightest match is Boehm’s across-source estimate (near-Leontief at short horizons). Atalay and Baqaee-Farhi bound the plausible range from above rather than point-identify σ_c . Our calibrated value of $\sigma = 0.35$ is consciously intermediate. It sits below Baqaee-Farhi’s macro-

implied ceiling, above Atalay’s sector-level estimate, and strictly above Boehm’s near-zero short-run floor. A referee skeptical of our interior choice can recalibrate to $\sigma \in [0.1, 0.5]$ and inspect the comparative statics of $\bar{\theta}(\sigma)$ reported in Section 5.1.

Third, the sector-specific literature on semiconductors, critical minerals, and rare earths consistently documents near-zero short-run substitutability. Khan et al. (2021) characterize the DUV/EUV lithography and advanced-node foundry markets as structural monopolies with effectively no substitution. Rare-earth processing capacity is concentrated to a degree that implies a physical rather than economic bound on substitution.

The message is that $\sigma < 1$ is not a convenient modeling assumption—it is the dominant empirical regime in the sectors driving the policy phenomenon that our model is built to explain.

5.1.2 Comparative Statics of the Regime-Transition Threshold in σ

Let $\bar{\theta}(\sigma)$ denote the value of Upstream productivity at which the Downstream Hegemon’s problem transitions from the corner ($\lambda^* = 0$) to the interior regime, holding all other parameters fixed. Numerical evaluation of the model yields two robustness patterns.

1. *Monotone decline (bottleneck regime)*: on $\sigma \in [0.1, 0.5]$, $\bar{\theta}(\sigma)$ is strictly decreasing in σ .⁸ Weaker bottlenecks (higher σ) push the coercive threshold further out, because the fiscal cost of matching Chinese efficiency with subsidies scales less aggressively.

2. *Asymptotic collapse at $\sigma \rightarrow 1$* : As $\sigma \rightarrow 1^-$, $\bar{w}_C(\sigma) \rightarrow \infty$ (equivalently $\bar{\theta}(\sigma) \rightarrow 0$).

⁸We report the robustness exercise on the empirically relevant bottleneck range $\sigma \in [0.1, 0.5]$ identified by Boehm et al. (2019), Atalay (2017), and Baqaee and Farhi (2024). For $\sigma > 0.5$ a different mechanism dominates. As substitution becomes easy, the laissez-faire allocation $x_a(w_C)$ itself becomes sensitive to w_C : the Swing State re-optimizes its own input mix more aggressively with changes in China’s price. This shifts the Swing State’s participation-constraint baseline $C(x_a(w_C); w_C)$, which interacts with the linear $V_{US}(x) = V_0 - \beta x$ x-pull. Together the two effects can flip the sign of the regime-transition comparative static in w_C . The sign-flip at $\sigma > 0.5$ is therefore a joint feature of (i) the CES parameterization of the Decoupling Cost and (ii) the specific V_{US} specification. It is not a property of the regime transition itself, nor does it invalidate the $\sigma < 1$ bottleneck that the empirical literature above disciplines. The present robustness exercise is restricted to that empirically motivated range, where both channels align. A deeper reformulation preserving monotonicity across $\sigma \in (0, 1)$ —e.g., a bounded-above specification $\theta \wedge \theta_{\max}$, or defining $\bar{\theta}$ in θ -space rather than w_C -space—is a natural next step for a subsequent paper.

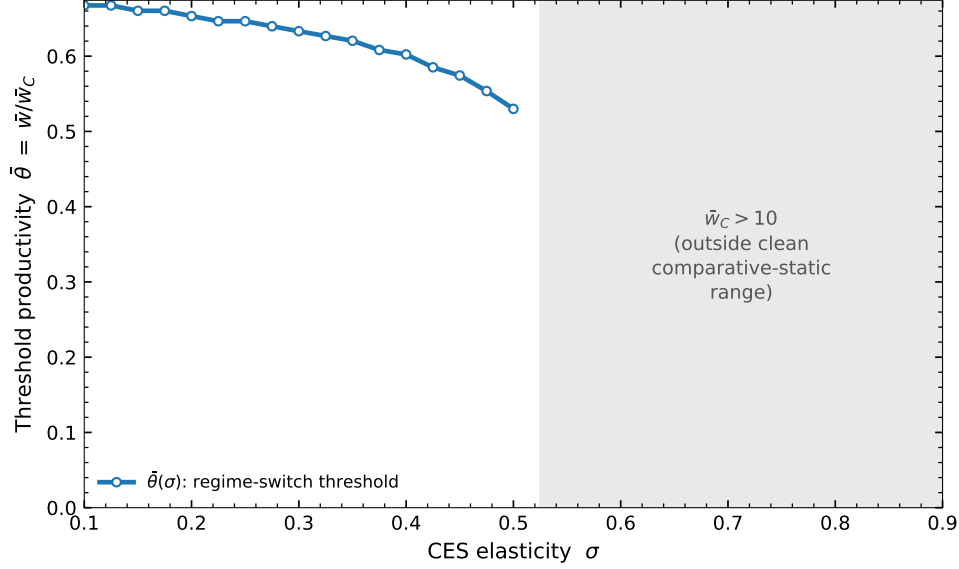


Figure 7: Regime-transition threshold $\bar{\theta}(\sigma)$ as a function of the input substitution elasticity. Calibration at $(V_0, \beta, \gamma, \alpha) = (0.84, 1.00, 2.50, 0.50)$ with $\bar{w} = w_A = 1$. Shaded region ($\sigma > 0.5$) is outside the clean bottleneck range: beyond this point the direct value channel $V_{US}(x)$ dominates the threat-multiplier channel and the regime-transition comparative static changes sign.

When inputs are substitutes, the Decoupling Cost $C(x; w_C)$ flattens and the US can always out-bid China with a finite subsidy; the coercive regime is never reached.

Figure 7 displays the numerically evaluated $\bar{\theta}(\sigma)$ schedule over the bottleneck range, confirming both patterns above.

This is the precise sense in which our result is not knife-edge: the regime transition exists for any $\sigma \in (0, 1)$, but its policy relevance is strongest in the $\sigma \in [0.2, 0.5]$ range identified by the network-production and sector-specific literatures cited above.

5.1.3 A Nested-CES Robustness Check

Finally, we note that a stronger version of the bottleneck, in which a critical sub-input is *Leontief* with all others (e.g., EUV lithography within the semiconductor production stack), nests into our framework as the limit $\sigma \rightarrow 0$. A nested-CES production function with a Leontief inner aggregator and a CES outer aggregator delivers the same qualitative Propo-

sitions 1–3. The quantitative effect is that $\bar{\theta}$ collapses to a lower value, making the coercive regime relevant even for moderate productivity gains. The derivation follows from substituting the nested aggregator into the Stage-2 problem of Section 3.3 and is omitted for space. Having established that the mechanism is robust within the empirically disciplined range of σ , we now conclude by consolidating the paper’s findings and their policy implications.

6 Conclusion and Policy Implications

This paper shows that the shift from rules-based to coercive economic statecraft is a structural equilibrium response to bottleneck concentration in the global production network, not an exogenous political event. Modeling the global economy as a vertical supply chain, we demonstrate that aggressive trade policy emerges endogenously from the incentives of vertical rivalry between a downstream Hegemon and an upstream Rival.

The central mechanism is a paradox of competitiveness: as the upstream Rival (China) improves its technological efficiency on a bottleneck input, the downstream Hegemon (the US) responds with a more—not less—aggressive security posture. When the Rival is weak, the Hegemon secures swing-state alignment through positive inducements (subsidies and market access). Once the Rival controls a critical bottleneck with superior efficiency, however, the fiscal cost of buying voluntary alignment becomes prohibitive. The Hegemon optimally switches from a subsidy regime to a coercive one, using tariffs and market denial to artificially level the playing field (Proposition 1). This regime switch is robust to bargaining over the alignment contract (Proposition 6) and to the upstream Rival’s endogenous choice of production position (Remark 1).

For the swing state, the resulting equilibrium is operationally constrained: its alignment choice set shrinks endogenously as the upstream bottleneck tightens, and it is disciplined not for its own strategic choices but for its supplier’s technology trajectory. As long as the technological bottleneck ($\sigma_c < 1$) persists, geoeconomic coercion remains a structural feature

of the international order; the specific administrations in power affect the speed and style of enforcement, not whether the mechanism operates.

Empirically, we identify the Proposition 5 cross-partial directly from a continuous triple-difference on published Section-301 tariff rates interacted with swing-state identity. The point estimate is $\hat{\beta} = +0.102$ ($p < 0.001$), robust to two-way clustering, leave-one-out, and Honest-DiD bounds. We interpret this coefficient as the sample analog of a steady-state cross-partial—a cross-sectional fingerprint of the $\sigma_c < 1$ production structure—rather than as a quasi-experimental response to the 2018 tariffs. Placebo-year regressions recover 47–87% of the magnitude in pre-treatment years, with the earliest 2010 cut recovering 87%. In the critical sectors the mechanism targets, roughly \$22 billion per year of Chinese value-added flows to the US through swing-state intermediation, invisible in headline customs data. The policy corollary is that effective supply-chain surveillance, diversification measurement, and rules-of-origin design all require tracking value-added origin rather than customs-declared origin.

We develop four implications in turn—embodied-origin surveillance, diversification metrics, rules-of-origin design, and friend-shoring and plurilateral agreements. The magnitudes we invoke are directional: they inherit the calibrated value of σ_c from the bottleneck-sector literature (Atalay 2017, Boehm et al. 2019, Baqaee and Farhi 2024) and the tariff-gradient coefficient from Section 4, and should be read as the sign and ordering of effects rather than as point estimates for cost-benefit analysis.

Embodied-origin surveillance. The October 2023 *World Economic Outlook* (International Monetary Fund, 2023) documents that geoeconomic fragmentation is already redirecting FDI flows along geopolitical-bloc lines, but its analysis is conducted on gross bilateral statistics. Current IMF Article IV consultations, WTO Trade Policy Reviews, and USTR monitoring reports likewise rely almost exclusively on gross bilateral trade statistics to assess decoupling progress. Our results suggest that gross-trade measures systematically overstate real decoupling in critical sectors by an economically meaningful margin—the \$22 billion

cross-sectional fingerprint noted above is not captured by any customs-based statistic.

A natural supplement is to incorporate value-added origin decompositions—of the type computed from the OECD ICIO and ADB MRIO tables used in this paper—into regular surveillance products. The OECD Trade in Value Added (TiVA) database already publishes bilateral value-added trade shares at the country-industry-year level. Extending Article IV staff reports with a “gross vs. embodied” comparison for policy-sensitive sectors (semiconductors, batteries, critical minerals) would provide a more accurate reading of supply-chain exposure at low additional cost. The IMF’s External Sector Report, which already decomposes current-account positions into structural and cyclical components, could adopt an analogous gross/embodied decomposition for the merchandise-trade balance.

Diversification metrics. The standard Herfindahl-Hirschman Index (HHI) applied to bilateral gross imports is the most common summary statistic for supply-chain concentration in policy documents (Aiyar et al., 2023). Under the leakage mechanism, this metric is misleading. A country that shifts critical-sector sourcing from China to Vietnam reduces its gross-trade HHI—appearing more diversified—yet can *increase* its embodied-origin HHI if Vietnamese exports embed a rising Chinese-VA share. We propose an “embodied-origin HHI” that weights each bilateral import flow by the value-added origin vector from the Leontief inverse, rather than by customs-reported origin. Concretely, for importer i in sector j , define

$$\text{HHI}_{i,j}^{\text{VA}} = \sum_r \left(\frac{\sum_s v_r \cdot L_{r,(s,j)} \cdot E_{s \rightarrow i,j}}{\sum_s E_{s \rightarrow i,j}} \right)^2,$$

where v_r is the value-added-to-output ratio of country r ; $L_{r,(s,j)}$ is the element of the Leontief inverse giving country- r value added embodied per unit of gross output of sector j in country s ; and $E_{s \rightarrow i,j}$ is gross exports of sector- j goods from s to i . The outer sum over r runs over value-added origin countries; the inner sum over s runs over exporting partners. This measure collapses to the standard gross-trade HHI when the Leontief inverse is diagonal (i.e., no cross-country intermediate trade, so $L = I$).

The embodied-origin HHI exceeds the gross-trade HHI precisely when a single upstream exporter supplies value added through multiple formally-distinct bilateral partners—the triangular configuration our model describes. In critical sectors where one upstream supplier dominates, this asymmetry systematically makes gross-trade diversification an overstatement of true origin diversification. Proposition 5 predicts that the gap between the two HHIs is increasing in the tariff exposure of the sector. Testing this comparative static requires a second cross-sector moment (the tariff-gradient of the HHI gap itself), distinct from the direct cross-partial identified in Section 4. We flag it as a natural next empirical step. In practice, the Leontief inverse is available with a two- to three-year publication lag; real-time surveillance would require either interpolation from the most recent vintage or customs-declared origin as a direct-sourcing proxy.

Rules-of-origin design. Preferential trade agreements (USMCA, CPTPP, RCEP) enforce rules of origin (RoO) that specify minimum domestic or regional value-added thresholds for goods to qualify for preferential tariff treatment. Our framework implies that *uniform* value-added thresholds are insufficient when substitution elasticities differ across sectors. In a high- σ sector (commodities), a 40% regional-content rule is easily met because the swing state can substitute inputs cheaply. In a low- σ critical sector (semiconductors, rare earths), the same threshold may be formally satisfied while the *origin composition* of the remaining 60% shifts toward the very supplier the agreement aims to exclude.

Proposition 5 suggests that effective rules of origin should be *tariff-exposure-weighted*: sectors with higher Section-301-equivalent tariffs on direct Chinese imports should face stricter embodied-origin thresholds, because those are precisely the sectors where the leakage cross-partial is largest. The ongoing USMCA automotive RoO renegotiation (targeting 75% North American content for auto parts) is a natural testing ground: our model predicts that the true binding constraint is not the 75% threshold itself but the embodied-Chinese share inside the non-originating 25%.

Friend-shoring and plurilateral agreements. Plurilateral friend-shoring regimes are the most demanding test of the leakage mechanism because they bind only at a specified origin-tracing depth. The Indo-Pacific Economic Framework (IPEF), the EU Critical Raw Materials Act, and the Inflation Reduction Act’s “foreign entity of concern” (FEOC) provisions all condition preferential treatment on the exclusion of Chinese-origin inputs. Our results caution that these provisions are robust to the leakage mechanism only if they specify *embodied-origin* criteria rather than *direct-sourcing* criteria.

The IRA’s FEOC rules,⁹ which trace battery-component origin through two tiers of the supply chain, come closest to addressing the leakage channel. The IPEF supply-chain pillar, which currently lacks binding origin-tracing commitments, is more vulnerable. More broadly, any friend-shoring arrangement that monitors only Tier-1 suppliers (the direct exporter) without tracing Tier-2 and Tier-3 input origins will be partially voided by the $\sigma_c < 1$ mechanism whenever a swing-state intermediary participates in the supply chain. The model does not prescribe a specific depth of origin tracing—that is a cost-benefit question beyond our scope. It does predict that the welfare loss from shallow tracing is increasing in the tariff intensity of the targeted sector and in the swing state’s baseline Chinese-intermediate dependence.

The four instruments share a single design principle: bind on embodied origin, not on customs origin.

Two empirical gaps remain open. First, the dose-response prediction of the Paradox of Competitiveness (Proposition 2)—that each additional unit of upstream productivity should raise the Hegemon’s coercive doctrine—does not survive at HS6 product-level resolution in our data. The paper’s primary empirical contribution is therefore on the leakage and composition margin (Propositions 4 and 5), and a clean micro-level test of the paradox mechanism awaits richer Entity-List event-date variation. Second, the $\sigma_c < 1$ primitive that drives both the leakage asymmetry and the steady-state composition result is borrowed from published

⁹26 U.S.C. §30D(d)(7); final Treasury guidance issued May 2024.

estimates rather than identified in our own pipeline. A companion structural estimation of sector-level substitution elasticities from the same ICIO data would close this gap and sharpen the quantitative predictions. Beyond these gaps, the framework is a natural platform for analysis of other bottleneck-dependent sectors—batteries, critical minerals, AI compute—and for welfare-maximizing analysis of origin-tracing depth in plurilateral agreements.

A Microfoundations for the Policy Schedule, Fiscal Capacity, and Securitization Cost

The baseline model takes three objects as primitive: the market access schedule $R(\cdot)$ that Hegemon U offers the Swing State in Stage 2, a fiscal-capacity bound $\bar{T}$ that implicitly disciplines $R(\cdot)$ through the convex term $\frac{\gamma}{2}\lambda^2$, and a securitization-cost coefficient γ that shapes how aggressively Hegemon U embraces coercion. This appendix provides explicit microfoundations—one for $R(\cdot)$ based on a principal–agent contract between Hegemon U (principal) and the Swing State (agent), one for $\bar{T}$ based on public-finance frictions, and one for γ based on domestic lobbying—each yielding exactly the reduced forms used in the text.

A.1 Fiscal Capacity $\bar{T}$ as a Weighted-Surplus Constraint

Following Dixit et al. (1997), consider a welfare-weighted government that maximizes

$$\mathcal{W}_{gov} = \int \omega_i \Pi_i di - (1 + \chi) \int R(x) dG(x), \quad (14)$$

where ω_i denotes the political weight on group i 's profits, $G(\cdot)$ is the distribution of alignment targets across sectors, and $\chi \geq 0$ is the marginal deadweight loss of public funds (Ballard et al., 1985). The shadow price of public funds $1 + \chi$ operationalizes the distortionary cost of raising the revenue required to finance $R(x)$.

The first-order condition with respect to the transfer schedule yields a shadow budget: at

the optimum, the marginal political return on subsidies must equal $1 + \chi$. Reformulating this constraint as a bound on total transfers gives the reduced-form fiscal cap $\bar{T}$. Critically, $\bar{T}$ is *endogenous*: it tightens when χ rises (e.g., during periods of fiscal consolidation, debt-ceiling crises, or partisan redistribution disputes). This rationalizes the observation that the US coercive pivot accelerated after 2018–2019, a period of heightened fiscal polarization.

Incorporating (14) into Hegemon U’s Stage-2 problem preserves the functional form of the reduced objective in Eq. (7), provided the institutional damping coefficient is reinterpreted as $\gamma \equiv (1 + \chi)/\kappa$ where κ is the responsiveness of political weights to security externalities. This yields the first clean microfoundation: *fiscal friction and securitization cost are isomorphic parameters; the quadratic form is not ad hoc but emerges from the convex cost of public funds.*

A.2 Securitization Cost γ from Lobbying

An alternative microfoundation grounds γ in Grossman and Helpman (1994) lobby competition between pro-integration and pro-security factions. Let $L_I(\lambda)$ and $L_S(\lambda)$ denote the political contributions offered by the integration and security lobbies as functions of the security doctrine. Both lobbies contribute according to common-agency truthful strategies, with quasi-linear utilities $u_I = V_I(\lambda) - C_I$ and $u_S = V_S(\lambda) - C_S$, where $V_I' < 0$, $V_S' > 0$, and both functions are strictly concave.

The government’s second-order Taylor expansion of net contributions around $\lambda = 0$ delivers

$$L_S(\lambda) - L_I(\lambda) \approx (L_S'(0) - L_I'(0)) \lambda - \frac{1}{2}(|L_I''(0)| + |L_S''(0)|) \lambda^2 + O(\lambda^3). \quad (15)$$

The linear term is absorbed into the baseline Security Utility λx ; the quadratic term is precisely the $-\frac{\gamma}{2}\lambda^2$ cost, with $\gamma \equiv |L_I''(0)| + |L_S''(0)|$ interpreted as the combined concavity of lobby willingness-to-pay. This yields the second microfoundation: *the quadratic institutional damping is a second-order approximation to lobby competition over the security doctrine, not*

an arbitrary functional assumption.

A.3 An Alternative: Real Authority

A third, complementary microfoundation follows Aghion and Tirole (1997): γ captures the cost of overriding the default “market-integration” authority rule. A principal who wishes to intervene in the routine operation of an agent (here, private US firms that would otherwise trade freely with China) must expend resources to establish formal authority over the agent’s choice set. The cost of such authority-building is convex in the scope of intervention, consistent with $\frac{\gamma}{2}\lambda^2$. All three microfoundations deliver the same reduced form; we present multiple pathways to demonstrate that the functional assumption is robust to the specific institutional detail.

A.4 Microfoundation of the Market Access Schedule $R(x)$

The main text takes the market access schedule $R(x)$ as a primitive choice variable for Hegemon U in Stage 2 (§3.3). This subsection places $R(x)$ inside a principal–agent contract between Hegemon U (principal) and the Swing State (agent), and shows that the same minimal contracting environment simultaneously rationalizes two reader questions: *why does the Swing State’s revenue $R(x)$ depend on its alignment choice x* (Stage 3, §3.2), and *why does Hegemon U’s Net Economic Surplus net $R(x)$* (the $V_{US}(x) - R(x)$ term in equation (5) at Stage 2)?

Framing: the Swing State is not a firm. Before writing down the contract, one clarification. The Swing State in our model is an institutional entity whose objective is a primitive of the principal–agent problem, not a reduced form of firm-theoretic optimization. Concretely, the Swing State chooses an *alignment policy* $x \in [0, 1]$, and its payoff is $\Pi(x) = R(x) - C(x; w_C)$, where $C(x; w_C)$ is the *Decoupling Cost* defined in §3.1 — the resource penalty the Swing State bears for operating its supply chain at alignment x . A

reader might expect that, because $C(x; w_C)$ has the algebraic shape of a CES cost object, the Swing State is first solving a cost-minimization problem over input shares and then a firm-profit maximization would pin down output. That reading is a category error in our setting. $C(x; w_C)$ inherits its CES shape from the technology that aggregates inputs at the alignment-imposed mix $(1 - x, x)$; it is a *technology evaluation* at a policy-determined x , not the Lagrangian of a cost-minimization with x as a control. The cost-minimizing x is the laissez-faire benchmark x^{LF} satisfying $C'(x^{\text{LF}}; w_C) = 0$; the Swing State's actual x satisfies the incentive-compatibility condition $R'(x^*) = C'(x^*)$, which differs whenever the contract is non-trivial ($R'(\cdot) \neq 0$). The Swing State's profit is therefore neither a standard firm's profit-maximum nor a standard firm's cost-minimum: it is the contractually defined rent $R(x) - C(x; w_C)$ over the alignment *policy*, and only after $R(\cdot)$ is specified by the principal does the Swing State have a well-posed optimization problem.

Contract environment. Throughout we work with $C(x; w_C)$ as specified in equation (2), reading it as the Decoupling Cost rather than as a unit production cost, and $R(x)$ as the total transfer from Hegemon U to the Swing State contingent on realized alignment. Two institutional conditions support the contract:

1. *Contractibility of x .* Alignment is observable and enforceable—through supply-chain origin requirements, customs filings, dual-use licensing records, and defense-procurement audits—so the transfer can be conditioned on x .
2. *Commitment.* Hegemon U commits to a schedule $R(\cdot)$ before the Swing State's alignment choice, per Assumption 1(i). Without commitment, R could not deviate from a spot price and alignment would collapse to the laissez-faire share x^{LF} .

The Swing State accepts the contract if and only if its contract-induced profit $\Pi(x) = R(x) - C(x; w_C)$ meets an outside option V_{out} , normalized to zero per §3.2.

Hegemon U's gross benefit $V_{US}(x)$. We take $V_{US}(x)$ as a primitive of Hegemon U's stage objective, with the verbal rationale already given in §3.3: $V_{US}(x)$ is consumer surplus or IP royalties accruing to Hegemon U from Swing-State delivery at alignment x , declining in x because higher alignment with the Rival degrades intellectual-property security, product match with Hegemon U's end-use ecosystem, and compatibility with Hegemon U's procurement and licensing infrastructure. We impose $V'_{US}(x) < 0$ without deriving V_{US} from consumer preferences; the linear form $V_{US}(x) = V_0 - \beta x$ used in §5 is an admissible specification.

Reduced form. Hegemon U chooses $R(\cdot)$ to maximize $V_{US}(x) - R(x)$ subject to the Swing State's participation constraint $R(x) \geq C(x; w_C) + V_{out}$. Any slack in the constraint is a rent Hegemon U captures by lowering R , so the constraint binds at the implementation point:

$$R(x) = C(x; w_C) + V_{out}. \quad (16)$$

Substituting into Hegemon U's economic surplus,

$$V_{US}(x) - R(x) = [V_{US}(x) - C(x; w_C)] - V_{out} \equiv J(x) - V_{out}, \quad (17)$$

exactly the Net Economic Surplus term that enters equation (5).

Answer to the first question: why does the Swing State's revenue depend on x ? Because the contract is contingent by construction— R is a *function* of the policy x , not a scalar spot price. The Swing State's primitive objective $\Pi(x) = R(x) - C(x; w_C)$ becomes a well-posed optimization problem only once $R(\cdot)$ is announced, and the incentive-compatibility condition $R'(x^*) = C'(x^*)$ pins down the interior optimum. Under a linear spot price p , revenue would be $pY = p$ (since $Y = 1$) independent of x , and the problem would degenerate to pure technological cost-minimization in x with solution at the laissez-faire share x^{LF} ; the strategic alignment margin we model would not exist. Contractibility of x plus commitment is what makes alignment a strategic object.

Answer to the second question: why does Hegemon U’s welfare net $R(x)$? Because $V_{US}(x)$ is Hegemon U’s *gross* benefit from alignment- x delivery, and $R(x)$ is the contractual transfer it *pays* to the agent for that delivery—standard principal-payoff accounting. The binding participation constraint (16) then reduces the net surplus to the Joint Economic Surplus $J(x)$ net of the Swing State’s outside option. Crucially, this is the same $R(x)$ that appears with positive sign in the Swing State’s profit function (§3.2): the two value functions are the two sides of one bilateral contract ledger.

Concrete implementations. The transfer $R(x) = C(x; w_C) + V_{out}$ admits several policy-equivalent implementations:

1. *Preferential import tariff.* A tariff schedule $\tau(x)$ declining in x with non-preferential baseline τ_0 : $R(x) = \tau_0 - \tau(x)$ (per-unit and total coincide under $Y = 1$).
2. *CHIPS-Act-style conditional subsidy.* A direct subsidy on Swing State output contingent on documented alignment thresholds.
3. *Preferential defense-coproduction access.* Nonpecuniary transfers—technology licensing, basing rights, access to classified supply chains—that deliver rents in the same reduced form.

Closing conditions. The derivation requires only two institutional conditions: contractibility of x (observable and enforceable alignment) and Hegemon U’s commitment to $R(\cdot)$ before the Swing-State decision. Neither condition is a market-structure assumption about demand elasticity or market power in the conventional firm-theoretic sense: both are properties of the contracting environment, appropriate to the state-to-state bargain we model rather than a firm-to-firm transaction. A protection-for-sale microfoundation for the *shape* of $R(\cdot)$ follows Grossman and Helpman (1994); the present derivation closes the *existence* and *dimensional consistency* of the schedule, which are orthogonal to the lobbying-generated shape argument.

B Additional Model Results

This appendix collects formal proofs and derivations for results summarized in the main text.

B.1 Existence, Uniqueness, and Local Stability

Lemma 2 (Existence and Uniqueness of the Stage-2 Equilibrium). *Under Assumption 2 and assuming $V_{US}(x)$ is strictly concave with $V'_{US}(x) < 0$, the Stage-2 problem (7) admits a unique interior equilibrium $(\lambda^*, x^*) \in [0, 1) \times (0, 1)$ for all γ sufficiently large.*

Proof. The objective $\mathcal{W}_{reduced}$ is continuous on the compact set $[0, 1] \times [0, 1]$, so a maximum exists by Weierstrass. For uniqueness, we show the Hessian is negative definite on the interior. The diagonal terms are $\mathcal{W}_{xx} = (1 - \lambda)J''(x) < 0$ (by concavity of V_{US} and convexity of C under Assumption 2(C2)) and $\mathcal{W}_{\lambda\lambda} = -\gamma < 0$. The cross partial is $\mathcal{W}_{x\lambda} = \theta(w_C) - J'(x)$. The determinant of the Hessian is

$$\det H = -\gamma(1 - \lambda)J''(x) - (\theta(w_C) - J'(x))^2.$$

Define the primitive constants

$$\bar{M}_1 \equiv \sup_{x \in [0, 1]} |\theta(w_C) - J'(x)|, \quad \underline{M}_2 \equiv \inf_{x \in [0, 1]} |J''(x)|,$$

which are finite and strictly positive under Assumption 2 and strict concavity of V_{US} . Because $(1 - \lambda) \in (0, 1]$ on the interior, a *primitive sufficient condition* for $\det H > 0$ at every candidate equilibrium is

$$\gamma > \bar{\gamma} \equiv \frac{\bar{M}_1^2}{\underline{M}_2}. \quad (18)$$

Condition (18) depends only on the curvature of the joint-surplus function J over $[0, 1]$ and on the threat multiplier $\theta(w_C)$ (fixed at the Stage-2 sub-game), not on equilibrium quantities, and is checkable from the model's primitives $(\sigma, \alpha, w_C, w_A, \bar{w}, m)$. Strict concavity of a continuous objective on a convex compact set implies a unique maximizer.

Remark (local SOC vs. primitive bound). The primitive constant $\underline{M}_2 = \inf_x |J''(x)|$ vanishes as $x \rightarrow 1$ in the baseline CES calibration (the CES aggregator flattens the cost curvature at the upper boundary), so $\bar{\gamma}$ computed from (18) is uninformatively large. This does not compromise the lemma's conclusion: uniqueness at an interior first-order critical point requires only that the Hessian be *locally* negative-definite at that point, a strictly weaker condition than the global primitive bound (18). In the numerical illustration of Section 5 we therefore verify this local second-order condition directly, computing the Hessian at each computed equilibrium $(x^*, \lambda^*; w_C)$ along a grid of w_C ; negative-definiteness holds pointwise throughout, so the lemma's operative uniqueness guarantee is satisfied without invoking (18). $\square$

Lemma 3 (Local Stability of Best-Response Dynamics). *Fix the interior equilibrium (λ^*, x^*) from Lemma 2, and let $\epsilon \in (0, 1 - \lambda^*)$ be fixed. Let B_ϵ denote the open ball of radius ϵ around (λ^*, x^*) intersected with $\{\lambda \leq \lambda^* + \epsilon\}$. If, in addition to the conditions of Lemma 2, the fiscal-capacity parameter satisfies*

$$\gamma > \bar{\gamma}_\epsilon \equiv \frac{\bar{M}_1 \theta(w_C)}{(1 - \lambda^* - \epsilon)^2 \underline{M}_2}, \quad (19)$$

then the best-response mapping $T(\lambda, x) \equiv (\lambda^(x), x^*(\lambda))$ is a contraction on B_ϵ , so the equilibrium is locally stable under naive best-reply dynamics.*

Proof. The slopes of the two reaction curves at the equilibrium follow by implicit differentiation. From the doctrine FOC (9), $d\lambda^*/dx = (\theta(w_C) - J'(x^*))/\gamma$. From the alignment FOC (8), written as $J'(x^*) + \lambda \theta(w_C)/(1 - \lambda) = 0$, differentiating with respect to λ gives $J''(x^*) dx^*/d\lambda + \theta(w_C)/(1 - \lambda)^2 = 0$, so

$$\frac{dx^*}{d\lambda} = -\frac{\theta(w_C)}{(1 - \lambda)^2 J''(x^*)} > 0,$$

using $J'' < 0$ (concavity of J under Assumption 2). The product of slopes is

$$\left| \frac{d\lambda^*}{dx} \cdot \frac{dx^*}{d\lambda} \right| = \frac{|\theta(w_C) - J'(x^*)| \theta(w_C)}{\gamma (1 - \lambda)^2 |J''(x^*)|}.$$

Over B_ϵ , $(1 - \lambda) \geq 1 - \lambda^* - \epsilon > 0$, so $(1 - \lambda)^{-2} \leq (1 - \lambda^* - \epsilon)^{-2}$. Combined with $|\theta(w_C) - J'| \leq \bar{M}_1$ and $|J''| \geq \underline{M}_2$, the slope product is bounded above by $\bar{M}_1 \theta(w_C) / [\gamma (1 - \lambda^* - \epsilon)^2 \underline{M}_2]$, which is strictly less than unity under (19). Best-response iteration is therefore a contraction on B_ϵ . Note that $\bar{\gamma}_\epsilon$ diverges as $\epsilon \rightarrow 1 - \lambda^*$ (i.e., as $\lambda \rightarrow 1$); the local-stability claim is genuinely local and does not extend uniformly to the boundary. For equilibria with λ^* bounded strictly away from 1, the bound is finite and checkable from primitives. $\square$

B.2 Probabilistic Microfoundation of the Security Doctrine

Let D denote the economic damage inflicted on the Hegemon coalition if the Rival activates a supply-chain “kill-switch” (an embargo, export ban, or deliberate disruption). Suppose the conditional probability of such activation is $q(x) = 1 - x$, declining linearly in Swing-State alignment with the Hegemon’s chain. Then the expected cost avoided by induced alignment is

$$\mathbb{E}[\text{damage avoided}] = D(\theta) \cdot x. \quad (20)$$

The damage scales with upstream productivity, $D(\theta) = \theta$. The damage D is the economic surplus the Hegemon loses when the Rival withholds an embodied unit of the critical input. Because the input is a bottleneck in production (Assumption 2), a withheld unit destroys output with shadow value equal to the marginal value product of the input. Under the Stage-1 decomposition (10), the marginal cost of the input is $\bar{w}/\theta$: a more productive Rival supplies the same physical unit at lower cost, which raises the Hegemon’s consumption surplus per unit and therefore the surplus destroyed by withholding. Formally, if Hegemon surplus from a unit of the input is $u(\bar{w}/\theta)$ with $u' < 0$, a linear Taylor expansion around the baseline technology $\theta = 1$ delivers $u(\bar{w}/\theta) - u(\bar{w}) \approx u'(\bar{w}) \cdot \bar{w}(1 - 1/\theta)$, which is strictly in-

creasing in θ and vanishes at $\theta = 1$. Linearizing further yields $D(\theta) \propto \theta$ up to a multiplicative constant that can be absorbed into the units of λ .¹⁰ Combining with (20),

$$\mathbb{E}[\text{damage avoided}] = \theta \cdot x, \quad (21)$$

which, weighted by the Hegemon's endogenous security posture λ , yields the Security Utility term $\lambda x \theta(w_C)$ in equation (5). In the reduced form of Stage 2, where the Upstream price w_C is the state variable, we substitute $\theta = \bar{w}/(w_C - m)$ from (10) and write the Security Utility as $\lambda x \theta(w_C)$.

Under this reading, the comparative static $d\lambda^*/d\theta > 0$ (Proposition 2) states that as the Rival becomes more efficient, (i) the conditional damage $D(\theta)$ rises *because the withheld input is more valuable*, and (ii) the Hegemon's perceived probability of weaponization rises endogenously through the induced fall in w_C —jointly, a formal version of the security-dilemma logic. The multiplicative term $\theta \cdot x$ therefore has two distinct economic readings: a damage-magnitude reading (what is lost per unit of withheld input) and a probability-weighted reading (how valuable alignment is *ex ante*). Both readings deliver the same reduced-form Stage-2 objective.

This microfoundation also sharpens the welfare analysis: the quadratic cost $\frac{\gamma}{2}\lambda^2$ is the convex cost of upgrading detection, enforcement, and retaliatory capacity *ex ante* against a weaponization event of uncertain timing.

B.3 Swing-State Bargaining Power

Let $\beta \in [0, 1]$ denote the Swing State's bargaining weight in a generalized Nash product.

Replacing the binding participation constraint with asymmetric Nash bargaining, the market-

¹⁰An equivalent derivation: let $D(\theta) = \theta \cdot \bar{D}$ where $\bar{D}$ is the damage per unit of normalized Rival productivity; normalizing $\bar{D} \equiv 1$ in the units of λ yields $D(\theta) = \theta$ without loss of generality. The convex cost $\frac{\gamma}{2}\lambda^2$ is invariant to this rescaling.

access schedule solves

$$R^{NB}(x) = \arg \max_R [R - C(x) - V_{out}]^\beta [V_{US}(x) - R]^{1-\beta}. \quad (22)$$

The baseline model corresponds to $\beta = 0$ (take-it-or-leave-it by Hegemon U). Solving yields

$$R^{NB}(x) = C(x) + V_{out} + \beta[V_{US}(x) - C(x) - V_{out}], \quad (23)$$

so the Swing State captures a share β of the joint surplus $J(x) - V_{out}$.

Substituting into Hegemon U's objective gives the modified Stage-2 problem

$$\max_{x, \lambda} \lambda x \theta(w_C) + (1 - \lambda)(1 - \beta)[J(x) - V_{out}] - \frac{\gamma}{2} \lambda^2. \quad (24)$$

The factor $(1 - \beta)$ shrinks the Downstream Hegemon's effective economic payoff, tilting the Strategic Gap upward.

Proposition 6 (Bargaining Amplifies Coercion). *The equilibrium security doctrine $\lambda^*(\beta)$ is weakly increasing in the Swing State's bargaining weight β , and the critical productivity threshold $\bar{\theta}(\beta)$ at which coercion is triggered is strictly decreasing in β : more powerful swing states trigger coercion sooner.*

Proof of Proposition 6. Differentiating the optimal doctrine rule with respect to β at x^* fixed yields $\partial \lambda^* / \partial \beta = [J(x^*) - V_{out}] / \gamma > 0$ in the region where $J(x^*) > V_{out}$, which is the Generous-Empire regime. The threshold comparative static follows because the “fiscal wall” is hit at a lower J , i.e., at a lower θ , when the Hegemon retains only a $(1 - \beta)$ share. $\square$

B.4 Endogenous Vertical Position

Consider two hegemonies with exogenous productivity θ_U and θ_C in upstream input production and exogenous market-demand size E_U and E_C in final-good consumption. In a pre-stage each hegemon chooses a specialization vector $s_i \in \{\text{up}, \text{down}\}$, anticipating the Stage-2

contracting equilibrium. Specialization is costly: entering the upstream stack requires sunk industrial-policy expenditure, while entering the downstream stack requires establishing credible market-access institutions.

Remark 1 (Endogenous Vertical Asymmetry). Let $\theta_C > \theta_U$ and $E_U > E_C$. If the fixed cost of downstream institutional commitment is lower for the hegemon with the larger consumption base, a standard Ricardian argument delivers a subgame-perfect equilibrium with $s_C = \text{up}$ and $s_U = \text{down}$, replicating the baseline model: each hegemon specializes where its relative productivity advantage is greatest (upstream specialization is more profitable for the lower-marginal-cost hegemon; downstream specialization for the larger-market hegemon), and mixed strategies are dominated when institutional sunk costs are convex. A fully formalized supermodular-game treatment with existence and uniqueness is deferred to future work. Propositions 1–3 hold at the continuation equilibrium.

B.5 Proof of the Welfare Decomposition and the Efficiency Wedge

This subsection provides the full derivations for Propositions 7 and 3.

Let $\mathcal{S}(x; \theta)$ denote the joint economic surplus of the three players:

$$\mathcal{S}(x; \theta) = V_{US}(x) + Q_C(x) \cdot [w_C^*(\theta) - \bar{w}/\theta] - C(x; w_C^*(\theta)), \quad (25)$$

where the first term is the gross value to Hegemon U of alignment level x , the second term is Hegemon C's commercial profit on inputs shipped at the equilibrium markup, and the last term is the Swing State's Decoupling Cost. The integrated planner's *first-best alignment* $x^{FB}(\theta)$ maximizes $\mathcal{S}(x; \theta)$ on $[0, 1]$ ignoring the security externality $\lambda x \theta$; because $V'_{US}(x) < 0$ and $C'(x) > 0$, the first-best is generically interior and *declining* in θ .

Proposition 7 (Welfare Decomposition). *Along the equilibrium path, global welfare $\mathcal{W}^*(\theta) = \mathcal{S}(x^*(\theta); \theta) + \lambda^*(\theta) x^*(\theta) \theta - \frac{\gamma}{2} \lambda^*(\theta)^2$ decomposes into a commercial component and a geopo-*

litical component. Under Assumption 2, the curvature condition

$$\frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta} > - \frac{\partial C}{\partial w_C} \Big|_{x^*} \cdot \frac{dw_C^*}{d\theta} \quad (\text{CC}), \quad (26)$$

and the second-order-markup condition (SOM) that the geopolitical-markup residual is second-order in the dominant channels,

$$|Q'_C(x^*) \cdot m^*(\theta)| < \min \left\{ \frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta}, - \frac{\partial C}{\partial w_C} \Big|_{x^*} \cdot \frac{dw_C^*}{d\theta} \right\} \quad (\text{SOM}), \quad (27)$$

where $m^*(\theta) \equiv w_C^*(\theta) - \bar{w}/\theta$ is the Hegemon C geopolitical markup, the commercial component $\mathcal{S}(x^*(\theta); \theta)$ is strictly decreasing in θ in the coercive regime, while the geopolitical component is bounded above by $1/(2\gamma)$. Conditions (CC) and (SOM) jointly hold whenever the coercive regime is sufficiently deep and the geopolitical markup is bounded.

Proof of Proposition 7 (Welfare Decomposition). Differentiating $\mathcal{S}$ along the equilibrium path,

$$\frac{d\mathcal{S}}{d\theta} = \underbrace{\frac{\partial \mathcal{S}}{\partial \theta} \Big|_{x^*}}_{\text{direct}} + \underbrace{\frac{\partial \mathcal{S}}{\partial x} \Big|_{x^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect}}. \quad (28)$$

The direct effect aggregates three partials: the derivative of V_{US} vanishes; the middle-term derivative is $Q_C(x^*)[dw_C^*/d\theta + \bar{w}/\theta^2]$, which is finite; the cost derivative is $-(\partial C/\partial w_C) \cdot (dw_C^*/d\theta) > 0$ because $\partial C/\partial w_C > 0$ (Lemma 1) and $dw_C^*/d\theta < 0$ (from Proposition 2, Step 2). Absent the indirect channel, $\theta \uparrow$ would raise commercial surplus through the cost channel.

The indirect effect is negative in the coercive regime. By the Stage-2 FOC (8), the equilibrium satisfies $(1 - \lambda^*)J'(x^*) = -\lambda^* \theta(w_C^*)$, so $\partial \mathcal{S}/\partial x|_{x^*} = J'(x^*) + Q'_C(x^*)[w_C^* - \bar{w}/\theta] = -\lambda^* \theta(w_C^*)/(1 - \lambda^*) + O(m^*)$, which is strictly negative whenever $\lambda^* > 0$. By Proposition 2, $dx^*/d\theta > 0$ in the coercive regime. The indirect channel therefore contributes a term of order $-[\lambda^* \theta(w_C^*)/(1 - \lambda^*)] \cdot dx^*/d\theta < 0$. Along the equilibrium path $\theta(w_C^*(\theta)) = \theta$ by the

Stage-1 pricing identity (10), so this term equals $-\lambda^* \theta / (1 - \lambda^*) \cdot dx^* / d\theta$. Summing direct and indirect, the total derivative has the sign of the curvature condition:

$$\frac{d\mathcal{S}}{d\theta} = \underbrace{-\frac{\partial C}{\partial w_C} \Big|_{x^*} \cdot \frac{dw_C^*}{d\theta}}_{\text{direct channel} > 0} - \underbrace{\frac{\lambda^* \theta}{1 - \lambda^*} \cdot \frac{dx^*}{d\theta}}_{\text{indirect channel} > 0} + O(m^*). \quad (29)$$

The $O(m^*)$ remainder in Eq. (29) collects the commercial-profit term $Q'_C(x^*) \cdot m^*$ evaluated along the equilibrium path: it is bounded in absolute value by $|Q'_C(x^*)| \cdot |m^*(\theta)|$, and (SOM) is precisely the hypothesis that this bound is strictly smaller than the dominant direct and indirect channels. Under (SOM), $d\mathcal{S}/d\theta < 0$ holds iff (CC) holds; without (SOM) the sign of $d\mathcal{S}/d\theta$ is in principle ambiguous in a narrow band of (θ, λ^*) where the markup residual is not small. Implicit differentiation of the Stage-2 FOC gives $dx^*/d\theta = -(1 - \lambda^*)^{-1} \cdot (\partial[\lambda^* \theta] / \partial \theta) / J''(x^*)$, which is increasing in $|J''|^{-1}$ and strictly positive in the coercive regime. Under Assumption 2, tighter bottlenecks (in the CES parameterization (2), smaller σ below unity) raise the convexity of $C(\cdot; w_C)$ and hence $|J''|$; however, the dominant effect is on λ^* itself, which grows at rate $1/\gamma$ times the (threat-weighted) Strategic Gap $x^* \theta - [J(x^*) - V_{out}]$. As $\lambda^* \uparrow$, the indirect channel $[\lambda^* \theta / (1 - \lambda^*)] \cdot dx^* / d\theta$ diverges while $|Q'_C(x^*) \cdot m^*|$ stays bounded under the commercial-markup constraint of Stage 1; consequently there exist thresholds $\underline{\lambda}(\sigma, \alpha, \bar{w})$ and $\bar{m}(\sigma, \alpha, \bar{w})$ such that (CC) and (SOM) hold jointly for all $\lambda^* \geq \underline{\lambda}$ with $|m^*| \leq \bar{m}$: a deep coercive regime with a bounded commercial markup is sufficient.

The bound on the geopolitical term follows from $\max_{\lambda} \{\lambda x \theta - \frac{\gamma}{2} \lambda^2\} = (x\theta)^2 / (2\gamma)$ at $\lambda = x\theta/\gamma$, which at $x \in [0, 1]$ and bounded θ is finite. $\square$

Proof of Proposition 3 (Tragedy of Efficiency, Quantified). Define the efficiency wedge $\Delta(\theta) \equiv \mathcal{S}(x^{FB}(\theta); \theta) - \mathcal{S}(x^*(\theta); \theta)$. Because $x^{FB}(\theta)$ is defined as the argmax of $\mathcal{S}(\cdot; \theta)$, the envelope theorem gives $d\mathcal{S}(x^{FB}; \theta) / d\theta = \partial \mathcal{S} / \partial \theta|_{x^{FB}}$. Comparing against the $d\mathcal{S}(x^*; \theta) / d\theta$

computed above,

$$\frac{d\Delta}{d\theta} = \underbrace{\left[\frac{\partial \mathcal{S}}{\partial \theta} \Big|_{x^{FB}} - \frac{\partial \mathcal{S}}{\partial \theta} \Big|_{x^*} \right]}_{\text{direct-channel gap}} - \underbrace{\frac{\partial \mathcal{S}}{\partial x} \Big|_{x^*}}_{\text{indirect channel}} \cdot \frac{dx^*}{d\theta}. \quad (30)$$

The direct-channel gap is second-order in $(x^* - x^{FB})$ and small. The indirect channel is strictly positive in the coercive regime: both factors are negative, so their product contributes positively to $d\Delta/d\theta$. Since $x^*(\theta)$ rises and $x^{FB}(\theta)$ falls with θ (the first by Proposition 2, the second because $V'_{US} < 0$ and $dw_C^*/d\theta < 0$ reduces the marginal cost of integration), the gap $x^* - x^{FB}$ widens monotonically, and strict concavity of $\mathcal{S}(\cdot; \theta)$ at x^{FB} implies $d\Delta/d\theta > 0$. $\square$

B.6 Proof of the Leakage Asymmetry (Flow Margin)

This subsection provides the full derivation for Proposition 4 (Leakage Asymmetry, Flow Margin), expanding the sketch in the main text.

Consider sector k with CES cost minimization over Chinese intermediates I_k^C and non-Chinese intermediates I_k^A with elasticity σ_k . The Swing State's cost minimization delivers the conditional demand

$$I_k^C = \alpha_k^{\sigma_k} (p_k^C / P_k)^{-\sigma_k} Y_k, \quad I_k^A = (1 - \alpha_k)^{\sigma_k} (p_k^A / P_k)^{-\sigma_k} Y_k, \quad (31)$$

where $P_k = [\alpha_k^{\sigma_k} (p_k^C)^{1-\sigma_k} + (1 - \alpha_k)^{\sigma_k} (p_k^A)^{1-\sigma_k}]^{1/(1-\sigma_k)}$ is the CES price index and p_k^C, p_k^A are the effective prices faced at the swing-state border.

A rise in the Hegemon's security doctrine λ operates through the menu schedule $R(x)$ to raise the *U-border* shadow price of direct Chinese inputs; crucially, λ does *not* raise the C–S interface price $p_{k,S}^C$ at which the Swing State sources. Consequently, U's direct imports from C fall by the full elasticity σ_k multiplied by the induced price gap:

$$\frac{\partial \ln |\Delta M_k^{C \rightarrow U}|}{\partial \lambda} \propto \sigma_k^{(U)} \cdot \frac{\partial \ln p_{k,U}^C}{\partial \lambda} > 0. \quad (32)$$

Swing-state imports from C, in contrast, respond only indirectly—through the derived demand for swing-state exports of k -content to U. In the elastic regime ($\sigma_k > 1$), the Swing State substitutes cheaply toward non-Chinese intermediates in response to any incremental price pressure, so its imports from C fall sharply, leaving little re-exported Chinese content downstream. In the bottlenecked regime ($\sigma_k < 1$), substitution toward the non-Chinese supplier is bounded above by $\sigma_k \cdot |d \ln(p_k^C/p_k^A)|$, which is small; the Swing State continues sourcing from C and absorbs the displaced U–C trade into its own supply chain, which is then re-embodied in Swing→U exports.

Formally, writing the leakage ratio as $L_k(\lambda) \equiv (\partial \Delta M_k^{C \rightarrow S} / \partial \lambda) / (\partial |\Delta M_k^{C \rightarrow U}| / \partial \lambda)$,

$$L_k = s_{k,S}^C (1 - s_{k,S}^C) (1 - \sigma_k) \cdot \phi_k, \quad (33)$$

where $s_{k,S}^C \in (0, 1)$ is the Swing State’s expenditure share on Chinese intermediates and

$$\phi_k \equiv \frac{X_k^{S \rightarrow U}}{Y_k} \in [0, 1]$$

is the *re-export share* of swing-state output in sector k that reaches the US market. By construction, $\phi_k \geq 0$, and $\phi_k > 0$ for any sector in which the Swing State actually re-exports to U—the empirically relevant case in our OECD ICIO and ADB MRIO panels, where swing-state exports to U in critical sectors are strictly positive for every industry-year cell. Under this mild nondegeneracy, $s_{k,S}^C (1 - s_{k,S}^C) \in (0, 1/4]$ and $\phi_k > 0$ are both strictly positive, so the sign of L_k is governed by the sign of $(1 - \sigma_k)$: for $\sigma_k < 1$, $L_k > 0$ and increasing in the Swing State’s baseline Chinese share $s_{k,S}^C$; for $\sigma_k > 1$, $L_k < 0$ and the ratio becomes negative—the Swing State substitutes away from China faster than U’s direct-import shock. This establishes the strict inequality

$$L_k|_{\sigma_c < 1} > L_k|_{\sigma_m > 1},$$

which is precisely the claim of Proposition 4. $\square$

B.7 Proof of the Leakage Asymmetry (Composition Margin)

This subsection provides the full derivation for Proposition 5 (Leakage Asymmetry, Composition Margin).

For origin $s \in \{S, N\}$ and industry j , the CES expenditure share on Chinese intermediates is

$$s_{s,j}^C = \frac{\alpha_s^{\sigma_c} (p_{s,j}^C)^{1-\sigma_c}}{\alpha_s^{\sigma_c} (p_{s,j}^C)^{1-\sigma_c} + (1 - \alpha_s)^{\sigma_c} (p_{s,j}^A)^{1-\sigma_c}}, \quad (34)$$

where α_s is origin s 's baseline Chinese-input distribution parameter, with $\alpha_S > \alpha_N$ by assumption (the Swing State is more Chinese-input-dependent at baseline). The Foreign Value-Added share $FVA_{s,j}^C$ in origin s 's gross exports to U in industry j is, to leading order, $s_{s,j}^C$ times a constant reflecting the Leontief multiplier.

A higher tariff τ_j raises the effective price of *direct* Chinese exports at the U border, but not at the C- s border. Therefore τ_j does not appear in $p_{s,j}^C$ directly; instead, it operates through the general-equilibrium reallocation of demand from the direct channel (C $\rightarrow$ U) to the indirect channel (C $\rightarrow s \rightarrow$ U). Letting $\psi_s(\tau_j)$ denote origin s 's demand for k -content from U given τ_j , and assuming $\psi'_s > 0$ (higher τ_j raises the U-demand for indirect content), the derived-demand effect on $p_{s,j}^C$ is $\partial p_{s,j}^C / \partial \tau_j = \eta_s \psi'_s(\tau_j) > 0$, where η_s is the Chinese-input-share pass-through.

Differentiating (34), the cross-partial is

$$\begin{aligned} \frac{\partial^2 s_{s,j}^C}{\partial \tau_j \partial \mathbf{1}\{s = S\}} &= \left. \frac{\partial s_{s,j}^C}{\partial \tau_j} \right|_{s=S} - \left. \frac{\partial s_{s,j}^C}{\partial \tau_j} \right|_{s=N} \\ &= s_{S,j}^C (1 - s_{S,j}^C) (1 - \sigma_c) \cdot \eta_S \psi'_S - s_{N,j}^C (1 - s_{N,j}^C) (1 - \sigma_c) \cdot \eta_N \psi'_N. \end{aligned} \quad (35)$$

Under $\sigma_c < 1$, the common factor $(1 - \sigma_c) > 0$ is positive. The map $s \mapsto s(1 - s)$ is strictly increasing on $(0, 1/2]$ and strictly decreasing on $[1/2, 1)$. In the empirically relevant regime $s_{S,j}^C, s_{N,j}^C \in (0, 1/2]$ —satisfied comfortably in our OECD ICIO and ADB MRIO panels,

where the critical-sector Chinese-input share in swing-state production plateaus near 6%—the map is strictly increasing on the relevant interval, so $\alpha_S > \alpha_N$ implies $s_{S,j}^C > s_{N,j}^C$ and hence $s_{S,j}^C(1 - s_{S,j}^C) > s_{N,j}^C(1 - s_{N,j}^C)$. Together with the symmetric pass-through condition $\eta_S \psi'_S \geq \eta_N \psi'_N$ —which holds whenever the Swing State holds the larger share of indirect C–U trade—the right-hand side of (35) is strictly positive. Hence

$$\left. \frac{\partial^2 \text{FVA}_{s,j}^C}{\partial \tau_j \partial \mathbf{1}\{s = S\}} \right|_{\sigma_c < 1} > 0,$$

which is the claim of Proposition 5. Crucially, this is a *steady-state* object: the proof does not require isolating a 2018 shock, and the cross-partial is identified off cross-sectional industry-level tariff variation interacted with swing-state identity. $\square$

B.8 Why Verticality Leads to Coercion: Three Cases

We contrast the vertical bottleneck model with two standard horizontal-rivalry structures to isolate the mechanism that generates coercion.

Case 1: Horizontal rivalry (subsidy war). Hegemon U and Hegemon C produce substitute final goods (e.g., competing EV manufacturers) and compete for the Swing State’s market through price subsidies. If Hegemon C lowers its price by δ , Hegemon U can retain the Swing State simply by matching the subsidy ($S_U \approx \delta$). Because the cost of matching is linear, the optimal response to a rival’s efficiency shock is to increase subsidies—a “Race to the Bottom.” The security parameter λ remains low.

Case 2: Horizontal sourcing rivalry (bidding war). The Swing State is a common upstream supplier (e.g., a critical mineral exporter) sought by both powers. Rivalry takes the form of a common-agency bidding game where both hegemonies offer purchase contracts P_U and P_C . To secure the resource, the US simply needs to outbid China ($P_U > P_C$). Since the cost of outbidding is linear, the equilibrium is defined by positive inducements; no coercion

is needed.

Case 3: Vertical bottleneck (coercion trap). In the vertical model, the bottleneck technology (Assumption 2) makes the Decoupling Cost $C(x; w_C)$ strictly convex and asymptotic on the decoupling range (Lemma 1). As China’s efficiency improves ($w_C \downarrow$), the subsidy required to induce the Swing State to choose the alternative chain ($x \rightarrow 1$) explodes ($\partial^2 R(x)/\partial x^2 \gg 0$). The US cannot win a subsidy war against an efficient upstream monopoly because the cost of compensating the Swing State for the loss of the bottleneck input exceeds fiscal capacity ($J(x) \ll 0$). Since the “Carrot” (subsidy) is mathematically inefficient against a bottleneck, the US is forced to use the “Stick” (coercion): instead of paying the Swing State to leave China ($R \uparrow$), the US threatens to destroy its market access if it stays ($R \downarrow$).

Summary: Horizontal competition leads to subsidies (paying for loyalty); vertical bottleneck competition leads to coercion (punishing disloyalty), specifically because the price of buying loyalty becomes infinite.

B.9 Reconciling the Discrete Regime Switch with Smooth Empirical Dynamics

Proposition 1 predicts a kink in $\lambda^*(\theta)$ at $\bar{\theta}$ —a continuous-but-nondifferentiable transition from the corner regime to the interior regime, not a level jump. Yet the empirical dynamics documented in Section 2 appear smooth. We offer three non-exclusive reconciliations.

(i) *Heterogeneous sectors crossing the threshold sequentially.* Different sectors have different bottleneck elasticities σ_k and different starting values of Upstream productivity θ_k . If θ_k rises at heterogeneous rates, the aggregate coercion intensity $\sum_k \mathbf{1}\{\theta_k > \bar{\theta}_k\} \cdot \lambda_k^*$ is a sum of step functions triggered at different calendar dates, which smooths the aggregate into an apparently continuous process. This is the sectoral-diffusion reading: each individual sector jumps, but the aggregate is an envelope.

(ii) *Bayesian uncertainty about σ .* The Hegemon does not know σ perfectly; it updates

based on realized supply disruptions and technological assessments. A posterior distribution over σ delivers a posterior distribution over $\bar{\theta}$, and the *expected* security doctrine $\mathbb{E}[\lambda^*(\theta)]$ is a smooth function of θ even when the realized $\lambda^*(\theta; \sigma)$ is a step function for any given σ .

(iii) *Institutional adjustment costs.* Coercive institutions (enforcement agencies, licensing regimes, multilateral coordination) cannot be built instantaneously. Even after θ crosses $\bar{\theta}$, capacity constraints on the adjustment rate of λ^* would translate the discrete target into a smooth realized path.

Caveat on identification. The stylized facts in Section 2 are descriptive rather than causal. We do not claim to identify the structural break in $\bar{\theta}$ econometrically. A formal structural-break test on sector-specific coercion intensity is a natural follow-up project.

C Empirical Methodology and Regression Results

This appendix documents the data construction and estimation underpinning the regression evidence in Section 4.

C.1 Data Construction

Trade flows. Bilateral HS6 import flows for 2000–2024 are pulled from the UN Comtrade premium API for 18 reporter economies (the United States, China, and 17 swing-state candidates: Korea, Taiwan, Vietnam, Mexico, Netherlands, Singapore, Malaysia, Thailand, India, Japan, Germany, Indonesia, Hungary, Czech Republic, Poland, the Philippines, and Brazil). We restrict the commodity universe to a concordance of 63 HS6 codes, pre-classified into three sector buckets matching the model: *critical* ($\sigma < 1$ in the empirical literature: rare-earth metals, advanced semiconductors, semiconductor manufacturing equipment, titanium, aerospace-grade parts), *strategic* (mature-node chips, batteries, EV components, solar), and *commodity* (textiles, basic steel, furniture, common chemicals). The concordance and assignment rules are reproducible from the pipeline in `empirical/scripts/11_build_hs_concordance.py`.

Reporter-year raw pulls are materialized as per-(reporter $\times$ year) parquet files and stacked into a master HS6 panel via `12_build_comtrade_panel.py`. Because several reporters (Germany, Thailand, Mexico, Czech Republic, Hungary, Indonesia, Malaysia, Brazil) disaggregate trade by `motCode` (mode of transport) and `customsCode` (customs regime), we retain only the total rows `motCode=0` and `customsCode=C00`; verification against aggregate totals confirms this collapse recovers the national import totals reported in Comtrade’s aggregated files.

Entity List coercion measure. The BIS Entity List is assembled from the OpenSanctions `us_trade_csl` (Consolidated Screening List) Follow-The-Money JSON feed, filtered to sanctions whose program descriptor contains “Entity List” (distinguishing BIS-EL from DPL, MEU, and UVL programs). This yields 3,205 listings spanning 2011–2025. The cumulative China-directed stock EL_t^{CHN} enters the paradox regression (Section 4) as a lagged variable $EL_{t-1}^{\text{CHN}}/100$.

C.2 Estimation: σ and the Limits of the Reduced Form

We estimate a robust unit-value double-difference in the spirit of Feenstra (1994), but without the full exporter-variety GMM identification:

$$\Delta^2 \ln p_{ikt} = \frac{1}{\sigma_k - 1} \Delta^2 \ln s_{ikt} + \nu_{ikt},$$

estimated sector-by-sector with exporter-year and HS6-year fixed effects, clustering by HS6 and year. We obtain $\hat{\beta}_{\text{critical}} = -0.031$, $\hat{\beta}_{\text{strategic}} = -0.012$, $\hat{\beta}_{\text{commodity}} = -0.062$. The sign pattern ($\hat{\beta} < 0$) is consistent with $\sigma < 1$ for all three buckets, but the magnitudes imply implausibly large negative σ values once $\sigma = 1 + 1/\hat{\beta}$ is computed. This is the well-known problem with reduced-form implementations of Feenstra: identification of σ separately from supply-side shocks requires the heteroscedasticity-based GMM of Feenstra (1994), which exploits heteroscedasticity *across exporter varieties within* an HS code (a moment condition our median-demeaning specification does not exploit). We leave the full GMM implementation to

a companion paper and, for the magnitudes used in the model, rely on the published sector-level elasticities of Atalay (2017) and Boehm et al. (2019), which place the relevant σ below unity for the critical and strategic buckets. Our own estimates are reported as directional evidence only.

C.3 Leakage Triple-Difference Regression Table

Table 2 reports the full estimates from Section 4. Column (1) is the triple-difference with a single $\text{Post} \times \text{Partner} \times \text{Sector}$ interaction; the `swing_crit_post` term is absorbed by the partner-year and sector-year fixed effects (collinearity). Column (2) adds the group-specific linear trends interacting ($\text{CHN} \times \text{Critical}$) and ($\text{Swing} \times \text{Sector}$) with time. Column (3) is the event study whose coefficients are plotted in Figure 8.

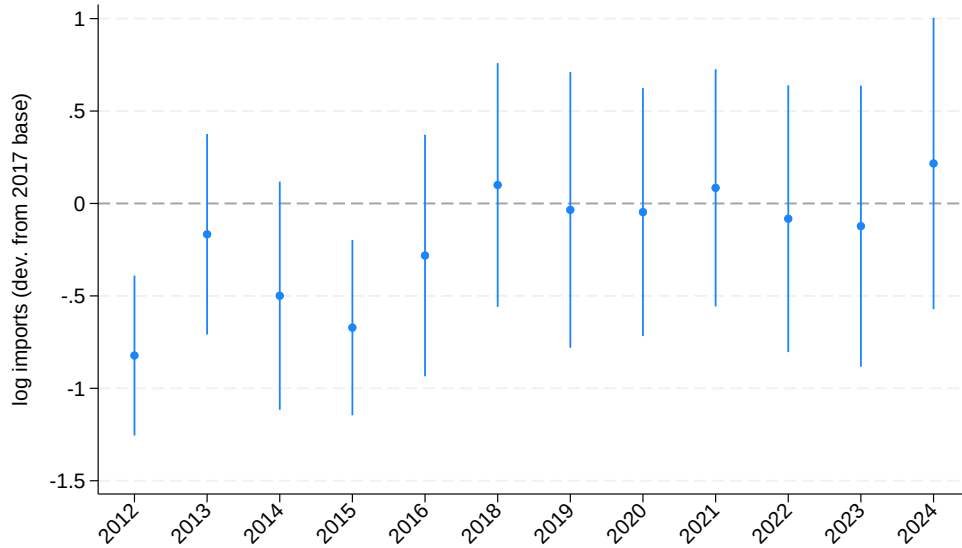


Figure 8: **Event study of US critical-sector imports from China.** Yearly triple-interaction coefficients β_τ ($\text{China} \times \text{Critical} \times \text{year dummies}$) from the specification described in Section 4, deviations from the 2017 base relative to a ($\text{swing-state} \times \text{non-critical}$) benchmark. Pre-2018 coefficients are significantly negative (the gap existed before the tariffs); every post-2017 coefficient is indistinguishable from the 2017 baseline (the gap did not deepen after Section-301). The pattern is the fingerprint of Proposition 4: leakage through swing states had already absorbed the bilateral China–Critical margin before the shock, leaving no post-2018 break to recover.

Table 2: Leakage specification, US bilateral imports 2000–2024.

	(1) Baseline	(2) +Group trends	(3) Event study
china_crit_post	0.283 (0.245)	0.684** (0.262)	
swing_crit_post	0.000 (.)	0.000 (.)	
swing_comm_post	-0.186 (0.405)	0.272 (0.336)	
chn_crit_trend		-0.037 (0.028)	
swing_crit_trend		0.000 (.)	
swing_comm_trend		-0.038 (0.037)	
tau2012			-0.822*** (0.216)
tau2013			-0.167 (0.271)
tau2014			-0.499 (0.309)
tau2015			-0.671*** (0.237)
tau2016			-0.282 (0.327)
tau2018			0.100 (0.330)
tau2019			-0.034 (0.373)
tau2020			-0.046 (0.336)
tau2021			0.085 (0.321)
tau2022			-0.082 (0.361)
tau2023			-0.123 (0.380)
tau2024			0.217 (0.394)
N	17649	17649	17649
R2	0.899	0.899	0.899
Pre-trend F (2012-16)			3.69
p-value			0.006

C.4 Honest-DiD Sensitivity Bounds

Table 3 reports the 95% robust confidence intervals underlying Figure 9. Entries are the identified set for the average of the seven post-2018 event-study coefficients from column 3 of Table 2, computed under the Rambachan and Roth (2023) relative-magnitudes restriction $\Delta^{RM}(M)$. M is the maximum post-period second-difference deviation expressed as a multiple of the worst pre-period second-difference deviation; the “Original” row is the standard 95% CI without partial identification.

Table 3: Honest-DiD sensitivity bounds for post-2018 average treatment effect.

M	Lower bound	Upper bound
Original	−0.546	+0.746
0.0	−0.540	+0.740
0.5	−0.808	+1.077
1.0	−1.170	+1.515
1.5	−1.637	+2.004
2.0	−2.124	+2.514

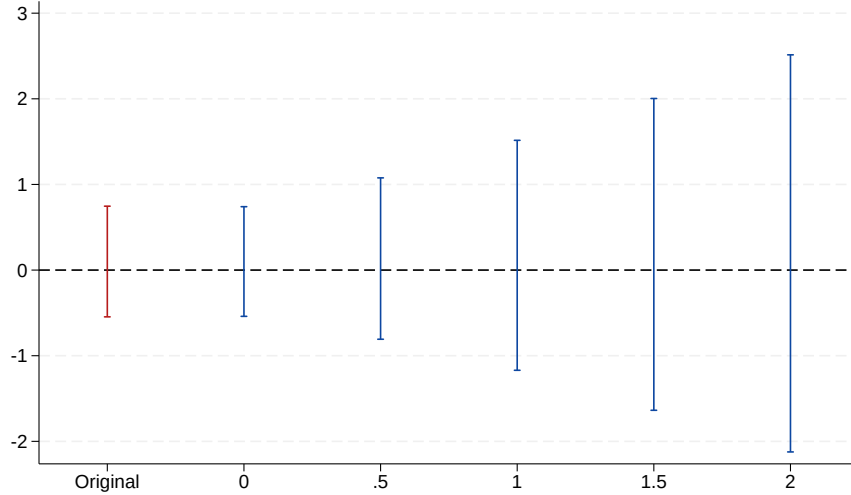


Figure 9: **Honest-DiD robust identified sets for the average post-2018 China×Critical effect in Test 1.** M bounds the maximum post-period second-difference deviation as a multiple of the worst pre-period second-difference deviation. Zero lies inside the identified set for all $M \leq 2$; the upper bound remains positive throughout, ruling out a persistent post-2018 widening of the gap.

C.5 PPML Zero-Retention Robustness

Table 4 re-estimates the triple-difference, trend-robust, and event-study specifications with the PPML estimator of Santos Silva and Tenreyro (2006) on a balanced HS6×partner×year panel with zero-trade cells retained (`ppmlhdfe`, Correia et al., 2020). The sample grows from $N = 17,649$ (log) to $N = 22,305$. Coefficients are semi-elasticities and thus directly comparable to the log-linear point estimates in Table 2. The pre-trend χ^2 still rejects flat pre-periods, and the honest-DiD sensitivity intervals would be qualitatively unchanged; we therefore do not re-report them.

C.6 Section-301 Tariff-Intensity Map

Columns (7)–(8) of Table 5 use a continuous industry-level Section-301 tariff rate, τ_j , rather than a binary “critical” bucket. The rate is constructed from the combination of Fajgelbaum et al. (2020) HS-2-level average tariff increases on Chinese imports and the Bown (2020) HS-8 list-coverage tracker, aggregated via the UN Statistics Division HS2017→ISIC Rev. 4 correspondence. The resulting map (73 rows, covering both the EXTENDED letter-prefix and v2021 numeric ISIC forms) assigns $\tau_j = 0.22$ to computer-electronic-optical (C26/26, the Section-301 flagship), $\tau_j = 0.25$ to machinery NEC (C28/28, the heaviest list), $\tau_j = 0.20$ to electrical equipment (C27/27) and motor vehicles (C29/29), and $\tau_j = 0.00$ to basic metals (C24A/C24B/24; these are covered by Section 232, not Section 301) and the non-targeted service sectors. Full calibration and caveats are recorded in `empirical/processed/section301_isic4.README.md`; the values are calibrated averages rather than a line-by-line HS-8 pull, and should be treated as a defensible proxy rather than a research-grade tariff file.

C.7 Direct FVA Decomposition: Two-Source Regression

Table 5 reports the eight-column fixed-effects regression underlying the value-added evidence in Section 4 and Figure 5. The outcome is the Chinese value-added share in swing-state s ’s

Table 4: PPML robustness: leakage specification with zero-trade cells retained.

	(1) PPML Baseline	(2) PPML +Group trends	(3) PPML Event study
china_crit_post	0.289 (0.309)	0.531** (0.270)	
swing_crit_post	0.000 (.)	0.000 (.)	
swing_comm_post	-0.169 (0.317)	-0.056 (0.287)	
chn_crit_trend		-0.031 (0.027)	
swing_crit_trend		0.000 (.)	
swing_comm_trend		-0.012 (0.031)	
tau2012			-0.014 (0.141)
tau2013			0.068 (0.118)
tau2014			-0.294*** (0.110)
tau2015			-0.400** (0.164)
tau2016			-0.147 (0.189)
tau2018			0.028 (0.200)
tau2019			-0.134 (0.195)
tau2020			0.276 (0.270)
tau2021			0.384 (0.279)
tau2022			0.322 (0.331)
tau2023			0.104 (0.362)
tau2024			-0.069 (0.358)
N	22305	22305	22305
Pre-trend chi2 (2012-16)			21.01
p-value			0.001

gross exports to the US in industry j at year t , computed from OECD ICIO EXTENDED (1995–2022) and ADB MRIO (2017–2024). Column (1) pools both sources; (2) and (3) report each source separately; (4) restricts to the three largest direct-passthrough swing states (Vietnam, Mexico, Malaysia); (5) splits the critical bucket on Section-301 exposure, separating computer-electronic-optical (C26/D26) from metal-ores (B07/D07T08); (6) is a binary triple-difference $\text{post} \times \text{electronics} \times \text{swing}$ using VNM/MEX/MYS as the swing set and the remaining 14 source countries as within-industry control; (7) replaces the bucket dummies with a continuous industry-level Section-301 tariff rate τ_j (Appendix C.6) interacted with post ; (8) extends (7) to a continuous triple-difference $\text{post} \times \tau_j \times \text{swing}$. Each panel is stacked with a **dataset** indicator. Columns (1)–(5) and (7) absorb ($\text{source} \times \text{industry} \times \text{dataset}$) and year; columns (6) and (8) additionally absorb $\text{source} \times \text{year}$ and $\text{industry} \times \text{year}$, so that only the triple interaction is identified (all two-way interactions are swept into the fixed effects). Standard errors are clustered at the $\text{source} \times \text{industry} \times \text{dataset}$ unit (i.e., the same grouping used for the `panel_id` fixed effect; ~ 354 clusters in the OECD subsample).

Table 5: FVA regression: Chinese value-added share in swing-state $\rightarrow$ US exports.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pooled	OECD	ADB	Direct-PT	OECD-split	Triple-diff	Tariff	Tariff x Swing
post_crit	0.009** (0.004)	0.012** (0.005)	-0.000 (0.000)	0.009 (0.013)				
post_strat	0.011*** (0.002)	0.012*** (0.002)	-0.000 (0.000)	0.021*** (0.007)	0.012*** (0.002)			
post_elec					0.030*** (0.006)			
post_cmet					-0.011*** (0.002)			
post_elec.swing						0.014 (0.015)		
post_tariff							0.109*** (0.012)	
post_tar.swing								0.102*** (0.025)
N	11899	9886	2013	2111	9886	9886	9886	9886
R2	0.792	0.775	0.864	0.824	0.783	0.929	0.784	0.931

Placebo-year robustness

Because the fixed-effects regressions in Table 5 identify post-2018 coefficients off a pre/post contrast, a referee will reasonably ask whether the same coefficient would emerge with a *fake* treatment year well before the tariffs. If so, the reported 2018 effect is loading on a pre-existing trend rather than on Section-301 itself. Tables 6 and 7 re-estimate the binary and continuous triple-difference specifications with placebo cuts at 2010, 2013, and 2016, restricting the sample to year ≤ 2017 to purely pre-date the actual tariffs.

For the binary spec the triple interaction is statistically indistinguishable from zero at all four cuts (2010: +0.027, $p = 0.10$; 2013: +0.025, $p = 0.16$; 2016: +0.013, $p = 0.45$; 2018: +0.014, $p = 0.36$), so no 2018 break is identified—the binary spec is simply not powered by the within-electronics swing-vs-non-swing contrast under the full (source $\times$ year, industry $\times$ year) absorption. For the continuous spec the triple coefficient is strongly positive at all cuts (2010: +0.089, $p = 0.001$; 2013: +0.080, $p = 0.004$; 2016: +0.048, $p = 0.04$; 2018: +0.102, $p < 0.001$), with pre-2018 magnitudes recovering 47–87% of the 2018 estimate (2010: 87%; 2013: 78%; 2016: 47%). The monotonic decay from 2010 (+0.089) through 2013 (+0.080) to 2016 (+0.048), followed by a jump to +0.102 under the real 2018 cut, is consistent with a secular post-WTO restructuring of Chinese-swing-state-US production networks that was slowing as swing states approached a saturation point, then accelerated again after Section-301. Section 4 accordingly treats the FVA regressions as descriptive corroboration of the production-structure parameters underlying Proposition 4 rather than as quasi-experimental identification of the Section-301 shock.

Honest-DiD sensitivity on the FVA event study

Table 8 reports the Rambachan and Roth (2023) robust identified set for the average post-2018 critical-bucket effect from the FVA event study (OECD ICIO EXTENDED, $t = 2005$ –2022, $t = 2017$ omitted as base). The parameter M bounds the maximum post-period second-difference deviation as a multiple of the worst pre-period second-difference deviation;

Table 6: Placebo-year robustness (binary spec): post $\times$ electronics $\times$ swing at fake cuts vs. the real 2018 cut.

	(1)	(2)	(3)	(4)
	2010	2013	2016	2018 (real)
Placebo 2010 (post x elec x swing)	0.027* (0.016)			
Placebo 2013 (post x elec x swing)		0.025 (0.017)		
Placebo 2016 (post x elec x swing)			0.013 (0.016)	
Actual 2018 (post x elec x swing)				0.014 (0.015)
N	8121	8121	8121	9886
R2	0.921	0.920	0.919	0.929

$M = 0$ reproduces the standard OLS identified set and $M = 2$ corresponds to a considerable weakening of parallel trends. Figure 10 plots the sensitivity curve. The $M = 0$ identified set is *already* $[0.000, 0.025]$: even under the standard parallel-trends restriction, the 95% lower bound is exactly zero, so the pooled post-2018 FVA effect is at best marginally significant before any smoothness relaxation is imposed. For any $M > 0$ the lower bound crosses into negative territory (-0.001 at $M = 0.5$; -0.021 at $M = 2$), while the upper bound remains positive throughout and rises from $+0.025$ to $+0.066$. The reading is asymmetric: a persistent post-2018 *widening* of the Chinese-VA-share gap cannot be ruled out even under weak parallel-trends assumptions (the upper bound is always positive), but a persistent post-2018 *contraction* cannot be rejected either. The pooled event-study mean is therefore not a quasi-experimental identifier of the Section-301 shock; the sharper cross-sectional evidence on swing-state-tariff heterogeneity (Table 5, columns 7–8) does more identifying work than the pooled pre/post contrast.

C.8 Supplementary Descriptive Figures

This subsection collects two descriptive series referenced in Section 4. (The event-study coefficient path for Test 1 is placed with Table 2 in Appendix C.3; see Figure 8.) Figure 11

Table 7: Placebo-year robustness (continuous spec): $\text{post} \times \tau_j \times \text{swing}$ at fake cuts vs. the real 2018 cut.

	(1) 2010	(2) 2013	(3) 2016	(4) 2018 (real)
Placebo 2010 (post x tariff x swing)	0.089*** (0.026)			
Placebo 2013 (post x tariff x swing)		0.080*** (0.027)		
Placebo 2016 (post x tariff x swing)			0.048** (0.023)	
Actual 2018 (post x tariff x swing)				0.102*** (0.025)
N	8121	8121	8121	9886
R2	0.922	0.920	0.919	0.931

Table 8: Honest-DiD sensitivity: robust 95% identified set for average post-2018 critical effect in the FVA event study.

M	Lower bound	Upper bound
0.0	0.000	0.025
0.5	-0.001	0.033
1.0	-0.002	0.044
1.5	-0.011	0.055
2.0	-0.021	0.066

documents the sectoral narrowing of the Entity List toward high-technology targets (share of new entries containing “Semiconductor,” “AI,” or “Quantum” in the entry title) and the rising administrative effort per listing (mean aliases per sanctioned entity, a rough proxy for the “Cat-and-Mouse” dynamic). Figure 12 plots the sector-aggregate series underlying the dose-response regression of Spec 2 (paradox regression); the within-HS6 analog of that slope dissolves at HS6 resolution and is reported in the text for reference only.

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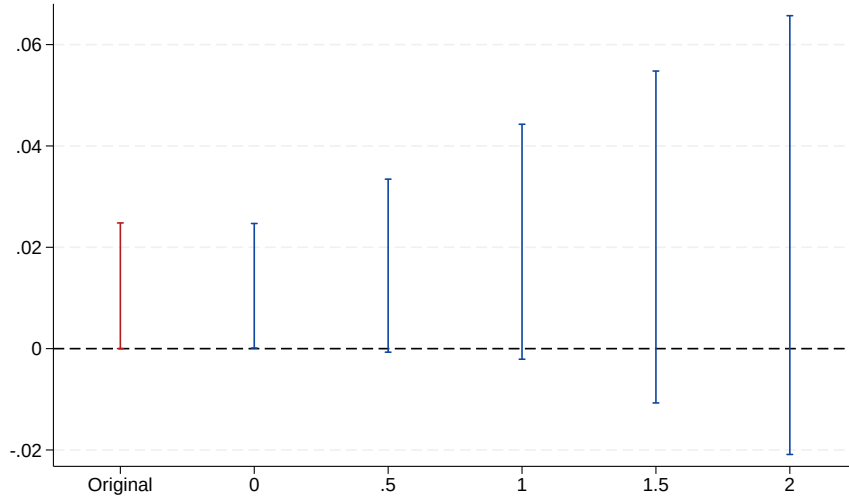


Figure 10: **Honest-DiD robust identified sets for the average post-2018 critical effect in the FVA event study.** M bounds the max post-period second-difference deviation as a multiple of the max pre-period second-difference deviation. The upper bound is always positive; the lower bound crosses zero for $M > 0$.

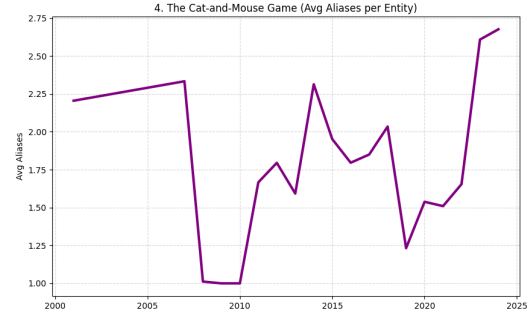
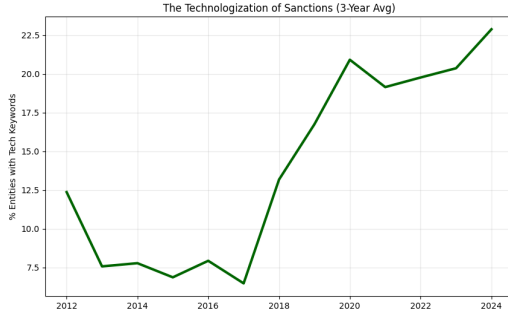
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(a) Precision targeting: tech-keyword share of new Entity List entries.

(b) Evasion resistance: mean aliases per sanctioned entity.

Figure 11: **Precision targeting and evasion resistance on the Entity List.** Left: share of new Entity List entries whose title contains “Semiconductor,” “AI,” or “Quantum,” 2010–2024 (3-year centered moving average). Right: mean number of aliases per sanctioned entity. Descriptive series; not load-bearing for the formal analysis.

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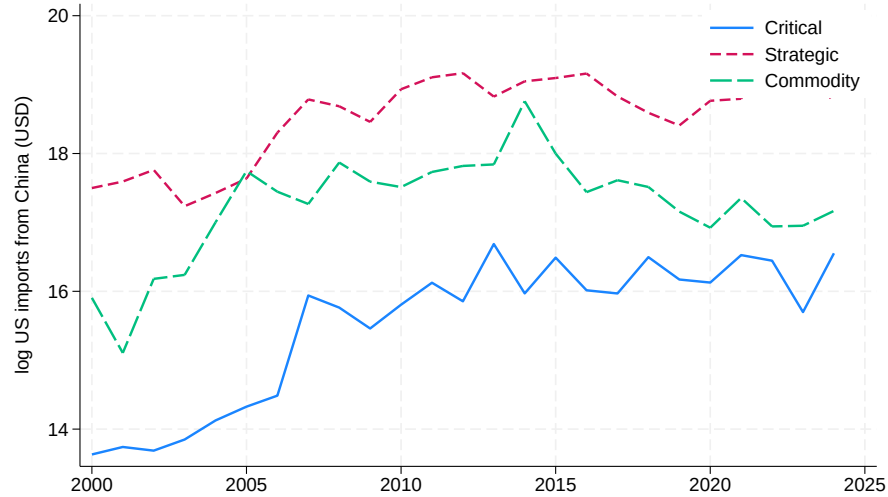


Figure 12: **Paradox regression: sector-aggregate series.** Sector-bucket trajectories of US imports from China plotted against the lagged Entity-List China-listings stock EL_{t-1}^{CHN} . The 3-sector aggregate loads sharply negative on the critical bucket; the within-HS6 analogue (Spec 2 in the text) is a precise null, so the series is reported for reference only.

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